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AI-Driven Knowledge Retrieval for Enterprise Architecture: A Design Science Study in an Industrial Context

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Abstract:

Enterprise Architecture (EA) teams manage large volumes of organisational knowledge, yet accessing and reusing this knowledge remains a persistent challenge due to fragmented repositories, inconsistent documentation practices, and the dependence on a small group of experts. This study investigates how Artificial Intelligence can improve EA knowledge accessibility through the design and evaluation of an AI-powered copilot within an industrial context. The research was conducted at Sandvik Coromant using a Design Science Research approach consisting of four phases: demand discovery and use case definition, technical evaluation, pilot realisation, and continuous learning. Data were collected through document analysis, twelve stakeholder interviews involving Enterprise Architects, Business Architects, and IT specialists, and a collaborative workshop to identify and prioritise AI-enabled EA use cases. The findings revealed a strong demand for faster knowledge retrieval, improved onboarding, support for Start Assurance activities, and democratisation of architectural knowledge across the organisation. The study compared Copilot Studio, custom RAG in Microsoft Foundry, and an open-source LLM approach, while also exploring vector and graph-based retrieval as possible extensions. Based on the results, an AI copilot prototype was developed to support the Start Assurance process by providing conversational access to EA knowledge and guidance throughout self-assessment activities. The formative evaluation indicated that the prototype was relevant for bounded first-line EA support, particularly for document discovery and guided preparation of use cases. The evaluation also confirmed the continued importance of source quality, governance, and human oversight. The study contributes a practical framework for identifying, prioritising, and implementing AI use cases within Enterprise Architecture, provides empirical insights into the adoption of generative AI for organisational knowledge management, and offers recommendations for future development through advanced retrieval mechanisms such as knowledge graphs and GraphRAG.

Keywords: Enterprise Architecture, Generative AI, Knowledge Management, AI Copilot, Design Science Research, Retrieval-Augmented Generation, Knowledge Graphs, Digital Transformation.

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1. Introduction

1.1 Background

Organisations operating in complex industrial environments increasingly depend on digital capabilities and coordinated decision-making across business and technology domains. Enterprise Architecture (EA) provides an organising logic through which business processes and information technology infrastructure can be aligned with an organisation's operating model and strategic priorities (Ross, Weill and Robertson 2006). EA also offers principles, methods, models, and governance practices for managing relationships between organisational structures, business processes, information systems, and technological infrastructure (Lankhorst 2017; The Open Group 2022).

However, the practical value of EA depends on more than the production of architecture models and formal documentation. Architectural principles, guardrails, capability models, roadmaps, and decision records must be understood and used by stakeholders involved in organisational change. This shifts attention from EA as a collection of artefacts towards EA as an organisational practice through which architects engage with stakeholders, communicate architectural direction, and make that direction applicable to ongoing work. Research on EA practice similarly shows that architecture involves a range of organisational artefacts and stakeholder interactions rather than the application of a framework in isolation (Kotusev 2018).

This practice-oriented perspective is also consistent with knowledge-management theory. Organisational knowledge exists in both explicit and tacit forms. Explicit knowledge can be represented in documents, procedures, models, and repositories, whereas tacit knowledge remains connected to individual experience and contextual judgement (Nonaka and Takeuchi 1995; Nonaka and von Krogh 2009). In an EA context, this distinction means that the availability of formal documentation does not necessarily ensure that stakeholders can interpret or apply it without support from experienced architects.

Recent research has explicitly positioned EA as a knowledge-management discipline and identified continuing issues such as knowledge fragmentation, loss of tacit knowledge, semantic ambiguity, tool constraints, and governance misalignment. The same research suggests that AI may support a transition from static EA documentation towards more dynamic knowledge infrastructure, while also identifying the need for further empirical investigation of AI-enabled EA practice (Toumi, Fosso Wamba and Hafsi 2025). This supports the relevance of examining EA knowledge access as an organisational problem rather than treating it solely as a document-retrieval issue.

The present study is situated at Sandvik Coromant, which forms part of Sandvik, a global high-technology engineering group providing solutions for the manufacturing, mining, and

infrastructure industries. According to the group's 2025 reporting, Sandvik employed approximately 42,000 people, generated revenues of approximately SEK 121 billion, and operated in more than 150 countries. Sandvik Coromant operates at the intersection of manufacturing tools, machining solutions, metal-cutting knowledge, technical services, software, and digital manufacturing support. This combination of industrial activity, specialist knowledge, and digital development provides a relevant setting for investigating how EA knowledge is made available across organisational boundaries.

Within the case organisation, the EA function supports the alignment of business objectives and technology through target architectures, transition planning, architecture-driven decisions, guardrails, and guidance for solution architects and product owners. The function is also involved in Start Architecture and Start Assurance activities that provide early framing and scoping for organisational initiatives. The internal thesis assignment further describes a need to move from static documentation towards an AI-supported service model capable of answering recurring questions, clarifying technical guardrails, and guiding employees through complex standards.

Developments in Large Language Models (LLMs), conversational AI, and Retrieval-Augmented Generation (RAG) create new possibilities for addressing this problem. RAG combines language generation with retrieval from external information sources, enabling responses to be grounded in knowledge retrieved at the time of interaction rather than relying exclusively on information stored in a model's parameters (Lewis et al. 2020). Recent systematic reviews nevertheless show that RAG introduces continuing challenges concerning retrieval quality, grounding, evaluation, and production-scale deployment (Brown, Roman and Devereux 2025; Karakurt and Akbulut 2026).

Against this background, the thesis investigates how an AI-supported EA guidance artefact can be designed and evaluated within an industrial setting. The study combines an organisational problem focus with the development of a purposeful artefact. It therefore examines not only the technical possibilities of AI-supported retrieval but also the stakeholder needs, knowledge conditions, governance arrangements, and design choices that influence whether such an artefact can provide useful organisational support.

1.2 Problem formulation

Enterprise Architecture is intended to support organisational coherence by connecting strategic objectives with business processes, information, applications, and technological infrastructure (Ross, Weill and Robertson 2006; Lankhorst 2017). However, the existence of EA models, principles, standards, and repositories does not by itself ensure that architectural knowledge influences practice. The value of this knowledge depends on whether stakeholders can identify relevant guidance and use it when making decisions.

This creates a particular difficulty in organisations where EA knowledge is distributed across documents, collaboration platforms, architecture repositories, presentations, process descriptions, and individual experts. Some of this knowledge is explicit and can be formally documented, while other elements depend on tacit knowledge developed through experience and professional judgement (Nonaka and Takeuchi 1995; Nonaka and von Krogh 2009). Consequently, stakeholders may locate a relevant document but remain uncertain about its authority, current status, or application to a particular situation.

The problem is therefore not simply a shortage of architectural documentation. It concerns access to usable and trustworthy EA knowledge. In the case organisation, the need for architectural support is associated with recurring questions, interpretation of technical guardrails, navigation of standards, and preparation for defined EA processes. The internal project description identifies the need for a first-line guidance capability that can improve access to this knowledge while preserving the role of the EA function.

AI-supported retrieval provides a possible response to this problem. RAG allows a language model to draw upon external sources when generating its response, which can improve the connection between generated content and domain-specific knowledge (Lewis et al. 2020). However, the implementation of such an approach does not automatically provide reliable organisational guidance. The quality of the response remains dependent on the relevance of the retrieved content, the condition of the knowledge sources, the retrieval configuration, and the controls governing how generated answers are interpreted.

The suitability of AI support is especially important in EA because architectural knowledge can influence organisational decisions concerning standards, technologies, capabilities, governance, and solution direction. Incorrect or insufficiently contextualised guidance may therefore create consequences beyond an unsuccessful document search. Recent research connecting EA, knowledge management, and AI similarly identifies fragmentation, tacit-knowledge loss, semantic ambiguity, and governance misalignment as continuing challenges, while positioning AI-enabled EA as a developing research direction rather than an established solution (Toumi, Fosso Wamba and Hafsi 2025).

The central problem addressed by this thesis is therefore how an AI-supported artefact can improve access to EA knowledge without presenting generated responses as a substitute for architectural judgement. Addressing this problem requires more than selecting an AI platform. It requires an understanding of existing knowledge-access challenges, stakeholder needs, source readiness, organisational governance, and the trade-offs between alternative solution architectures.

The study responds to this problem by examining the existing EA knowledge environment, deriving design requirements from empirical evidence, evaluating alternative AI approaches, and designing and evaluating an artefact within a selected pilot domain. In this way, the research treats AI-supported EA guidance as a socio-technical design problem involving technology, organisational knowledge, governance, and human expertise.

1.3 Research Gap

Enterprise Architecture has been widely studied as a means of supporting organisational alignment, governance, and coordinated change. Existing literature describes EA as an organising logic through which business processes, information systems, and technological infrastructure can be aligned with an organisation's strategic and operational priorities. Frameworks such as TOGAF provide structured methods, principles, and governance mechanisms for developing and maintaining architecture, while management-oriented perspectives emphasise the role of EA in creating coherence between business and information technology (The Open Group 2022; Ross, Weill and Robertson 2006). These contributions provide an established foundation for understanding the purpose, structure, and governance of Enterprise Architecture. However, they offer less detailed explanation of how architectural knowledge can be made continuously accessible to the wider population of employees who need to interpret and apply it in their everyday work.

This distinction is important because the practical value of Enterprise Architecture depends not only on the quality of architectural models, principles, standards, and governance structures, but also on whether these forms of knowledge can be located, understood, and applied by relevant stakeholders. Architecture knowledge may be formally available through documents, repositories, models, presentations, guardrails, and process descriptions, yet its availability does not necessarily ensure its accessibility. Stakeholders may still experience difficulties identifying the correct source, determining whether its content is current, interpreting abstract architectural guidance, or understanding how a principle applies to a specific organisational or technical situation. Consequently, a gap may exist between the knowledge formally produced through EA practice and the ability of employees to use that knowledge effectively.

Knowledge-management literature provides a complementary perspective by explaining how organisational knowledge is created, stored, shared, and reused. A central distinction in this field is made between explicit knowledge, which can be documented and transferred through formal artefacts, and tacit knowledge, which remains connected to individual experience, judgement, and contextual understanding. Nonaka and Takeuchi (1995) explain organisational knowledge creation as an interaction between tacit and explicit forms of knowledge. This perspective is particularly relevant to Enterprise Architecture because architectural guidance commonly exists in both forms. Principles, standards, models, and guardrails represent explicit knowledge, while the interpretation of these artefacts frequently depends on the experience of architects and other domain specialists.

Existing knowledge-management research therefore contributes important explanations of how knowledge is externalised, combined, shared, and internalised within organisations. Nevertheless, the literature does not fully resolve the practical problem of providing timely and contextual access to specialised knowledge in complex organisational settings. Conventional knowledge-management approaches often depend on repositories, document structures, search functions, communities of practice, or direct interaction with knowledgeable individuals. Although these mechanisms support knowledge sharing, they may be insufficient when information is distributed across several sources, expressed in different formats, or requires expert interpretation before it becomes practically useful. The challenge is therefore not limited to storing EA knowledge. It also concerns how employees can interact with that knowledge, receive explanations suited to their situation, and recognise when expert involvement remains necessary.

Recent developments in Artificial Intelligence, particularly Large Language Models and retrieval-augmented generation, introduce new possibilities for addressing this challenge. Large Language Models enable users to interact with information through natural-language questions and can produce explanations, summaries, and structured responses. However, their use in organisational settings introduces concerns regarding factual accuracy, outdated information, unsupported claims, and the generation of responses that are not grounded in approved internal sources. Retrieval-augmented generation addresses part of this problem by retrieving relevant content from external knowledge sources and supplying that content to a language model during response generation (Lewis et al. 2020). This approach provides a technical basis for connecting conversational AI with internal documents and knowledge repositories.

Research on RAG has consequently expanded the technical understanding of information retrieval, document segmentation, vector-based search, response generation, and grounding. More recent reviews of RAG and Large Language Models in enterprise knowledge management nevertheless indicate that a comprehensive understanding of their use in real organisational workflows remains limited and that many implementations continue to be experimental rather than fully integrated into production environments (Karakurt and Akbulut 2026). This suggests that technical progress in retrieval and language generation has not yet

been matched by an equivalent understanding of how these capabilities should be designed, governed, evaluated, and maintained within specific organisational practices.

A further limitation is that much of the existing AI literature is technology-centred. Studies commonly examine model performance, retrieval accuracy, response quality, system architecture, or general enterprise knowledge-management applications. These perspectives are necessary, but they do not sufficiently explain how an AI-supported artefact should be developed for Enterprise Architecture as a specialised organisational practice. EA knowledge is not limited to factual information that can be retrieved from documents. It includes principles, relationships, decision boundaries, capability information, governance expectations, architectural precedents, and guidance that may require interpretation according to the user's role and context. An AI solution designed for this environment must therefore support more than document discovery. It must also provide transparent, context-sensitive, and source-grounded guidance while preserving the authority of human architects in situations requiring professional judgement.

The existing literature also provides limited empirical insight into how different groups of EA stakeholders perceive the introduction of such support. Enterprise architects, solution architects, engineers, security specialists, integration specialists, product teams, and business stakeholders may interact with architectural knowledge in different ways. Some users may require support in locating documents, while others may need assistance interpreting guardrails, identifying capability ownership, understanding system relationships, preparing for architecture reviews, or determining when an issue should be escalated. A technically capable retrieval system may therefore remain organisationally ineffective if its design does not account for these differing needs, levels of expertise, and expectations regarding trust.

Similarly, limited guidance exists regarding how stakeholder needs should be translated into explicit design requirements for an AI-supported EA artefact. Decisions concerning approved knowledge sources, access control, response grounding, source citation, clarification, confidence, escalation, feedback, and content ownership are central to the practical usefulness of such an artefact. However, these concerns are often treated as implementation matters rather than as interconnected design requirements emerging from the organisational problem itself. This leaves an important methodological gap between identifying difficulties in accessing EA knowledge and designing an artefact that responds to those difficulties in a systematic and evaluable manner.

Design Science Research provides a suitable methodological foundation for addressing this gap because it connects the investigation of a relevant organisational problem with the design and evaluation of a purposeful artefact. The Design Science Research Methodology proposed by Peffers et al. (2007) provides a structured progression from problem identification and definition of solution objectives to design, demonstration, evaluation, and communication. Nevertheless, the application of this methodology to AI-supported Enterprise Architecture

knowledge guidance remains insufficiently examined. In particular, further research is needed to demonstrate how stakeholder evidence, organisational constraints, knowledge-source conditions, and alternative AI architectures can be combined to inform an artefact that is both technically feasible and organisationally relevant.

Taken together, the literature reveals three interconnected gaps. First, Enterprise Architecture research explains alignment, governance, and architectural practice but provides limited insight into how EA knowledge can be made directly accessible and understandable to diverse organisational stakeholders. Second, knowledge-management research explains the creation and sharing of explicit and tacit knowledge but offers limited guidance for providing interactive and context-sensitive access to specialised architectural knowledge distributed across repositories and individuals. Third, AI and RAG research demonstrates the technical potential of grounded knowledge retrieval but provides insufficient empirical and design-oriented guidance for developing trustworthy AI-supported EA artefacts within complex industrial organisations.

The central research gap addressed by this study therefore concerns the absence of a stakeholder-grounded and design-oriented account of how AI-supported guidance can be developed to improve access to Enterprise Architecture knowledge in an industrial context. More specifically, limited research explains how organisational needs can be translated into design requirements, how alternative AI solution approaches can be evaluated against contextual criteria, and how a suitable artefact can be designed to improve knowledge accessibility without replacing architectural judgement or formal governance.

This study addresses that gap through an industrial case at Sandvik Coromant. The study combines stakeholder interviews, internal document analysis, thematic analysis, solution evaluation, artefact design, and formative evaluation.

By addressing these dimensions together, the research moves beyond evaluating AI as a general search technology. It investigates AI-supported EA guidance as a socio-technical artefact whose usefulness depends on the relationship between technology, knowledge quality, stakeholder needs, governance structures, and human expertise. The intended contribution is therefore not limited to the selection of a particular platform or retrieval technique. Instead, the study seeks to provide a grounded design approach for developing and evaluating AI-supported access to Enterprise Architecture knowledge in a real organisational environment.

1.4 Purpose of the Study

The purpose of this thesis is to investigate how AI-supported guidance can improve access to Enterprise Architecture knowledge within an industrial context. The study examines existing knowledge-access challenges, identifies stakeholder needs and artefact requirements, evaluates alternative AI solution approaches, and designs and evaluates an AI-supported Enterprise Architecture guidance artefact within a selected pilot domain. In doing so, the thesis seeks to provide a practical contribution to the case organisation and a research contribution concerning the design of AI-supported guidance for knowledge-intensive Enterprise Architecture practice.

1.5 Research questions

The study is guided by the following research questions:

> RQ1: What challenges affect access to Enterprise Architecture knowledge in the case organisation?
>
> RQ2: What stakeholder needs and design requirements should inform an AI-supported Enterprise Architecture guidance artefact?
>
> RQ3: Which AI solution approach is most suitable for the selected pilot when evaluated against the identified organisational and technical criteria?
>
> RQ4: How can an AI-supported Enterprise Architecture guidance artefact be designed and evaluated within the case organisation?

1.6 Working hypothesis

This thesis is guided by a working hypothesis stating that, if Sandvik Coromant adopts a carefully evaluated AI approach selected through stakeholder-driven criteria and validated through a pilot domain, then an intelligent EA support artefact can improve access to architectural knowledge by reducing dependence on individual experts and enabling faster and more consistent guidance. This overall hypothesis is supported by three underlying assumptions. First, if EA documentation in the pilot area is sufficiently structured and prepared for retrieval, AI-supported retrieval should perform better than current manual search practices in terms of relevance and accessibility. Second, if multiple solution alternatives are evaluated using criteria such as accuracy, sovereignty, maintainability, and cost, a suitable solution approach can be identified for the Organisational context. Third, if the artefact includes a mechanism for feedback and knowledge updating, it can remain useful as architectural knowledge evolves over time.

1.7 Delimitations

This thesis is limited to one industrial case context, namely the Enterprise Architecture environment at Sandvik Coromant. The study does not attempt to generalise statistically to all Organisations but instead seeks to provide an analytically grounded understanding of one case and derive design implications from it. The work is further limited to the design and evaluation of an AI-supported guidance concept within a selected pilot domain rather than a full enterprise-wide implementation. In addition, the study focuses on knowledge access and guidance support, not on replacing architects or automating all architectural decision-making. Technical implementation is therefore treated at the level necessary to design, justify, and validate the artefact concept within the available thesis scope and time frame.

1.8 Thesis structure

The thesis is organised into nine chapters. Chapter 1 introduces the research background, problem formulation, research gap, purpose, research questions, working hypothesis, and delimitations. Chapter 2 reviews the literature on Enterprise Architecture, knowledge management, AI-supported knowledge retrieval, enterprise assistants, and Design Science Research, and presents the analytical framework. Chapter 3 describes the research methodology, including the research process, data collection, data analysis, artefact-design process, and evaluation approach. Chapter 4 presents the case organisation, the Enterprise Architecture function, relevant knowledge artefacts, and the existing support context. Chapter 5 reports the empirical findings, derives the design requirements, and explains the selection of Start Assurance as the pilot domain. Chapter 6 presents the candidate solution approaches, architecture selection, proposed artefact, prototype design, and scaling direction. Chapter 7 presents the formative evaluation and its results. Chapter 8 discusses the findings in relation to the research questions, working hypothesis, and literature. Chapter 9 concludes the thesis by presenting the answers to the research questions, contributions, limitations, and directions for future research.

2. Literature Review and Theoretical Foundation

2.1 Enterprise Architecture as an Organisational Practice

Enterprise Architecture is commonly presented as a discipline for coordinating an organisation's business objectives, processes, information, applications, and technological infrastructure. Ross, Weill and Robertson (2006) define EA as the organising logic for business processes and IT infrastructure, reflecting the integration and standardisation requirements of the organisation's operating model. This definition positions EA as a connection between organisational direction and the technological foundation through which that direction is implemented.

Lankhorst (2017) similarly describes EA as a coherent set of principles, methods, and models used in the design and realisation of an organisation's structures, processes, information systems, and infrastructure. This perspective emphasises the ability of EA to provide an integrated view of the organisation and the relationships between its constituent parts. Rather than focusing on a single system or project, EA seeks to maintain coherence across a changing organisational and technological environment.

However, defining EA primarily through principles, methods, and models may place excessive emphasis on its formal outputs. Architectural descriptions do not produce organisational value independently. They must be communicated, interpreted, and incorporated into decisions made by business leaders, architects, product owners, engineers, and other stakeholders. EA should therefore also be understood as an organisational practice involving interaction between people, artefacts, governance arrangements, and decision processes.

Kotusev (2018) examines this distinction through an exploratory case study of TOGAF-based EA practice. The study shows that organisations may adopt selected EA artefacts and practices while adapting or disregarding substantial elements of formal frameworks. The case used company-specific terminology, governance arrangements, maturity models, and adapted reference models rather than implementing TOGAF as a complete method. This suggests that EA practice is shaped by organisational needs and established ways of working rather than determined solely by compliance with a formal framework.

Kotusev (2018) also describes EA as a set of descriptions intended to bridge communication between business and IT stakeholders, facilitate information-systems planning, and improve alignment. This practice-oriented interpretation is particularly relevant to the present study because the research problem concerns whether architectural knowledge can be used by stakeholders beyond the EA function.

The value of EA therefore depends on the relationship between formal artefacts and organisational action. Principles, standards, guardrails, capability maps, target architectures, and decision records support practice when stakeholders can identify the relevant material and understand its implications. If these connections are weak, an organisation may possess extensive architecture documentation while continuing to depend on individual architects to explain what the documentation means.

Recent research reinforces this interpretation by positioning EA as a knowledge-management discipline. Toumi, Fosso Wamba and Hafsi (2025) connect EA artefacts and governance practices with the capture, formalisation, sharing, and reuse of organisational knowledge. The authors identify knowledge fragmentation, tacit-knowledge loss, semantic underspecification, tool constraints, and governance misalignment as continuing barriers within EA practice. This provides a direct theoretical basis for examining architectural knowledge access as both an EA problem and a knowledge-management problem.

2.1.1 TOGAF and Enterprise Architecture Practice

The Open Group Architecture Framework is one of the established standards for Enterprise Architecture. The TOGAF Standard provides concepts, methods, and guidance for developing, governing, and maintaining EA practices. It distinguishes core concepts and practices from supporting guides that organisations can configure according to their circumstances (The Open Group 2022).

The Architecture Development Method forms a central part of TOGAF. It provides an iterative process for examining an organisation's current architecture, defining a desired future architecture, assessing gaps, and planning the transition between existing and intended states. Its scope covers business, data, application, and technology concerns and includes attention to stakeholder engagement, requirements management, implementation governance, and architecture change.

TOGAF also recognises the importance of architecture artefacts. Principles, standards, models, roadmaps, reference architectures, and decision-related documents provide representations through which architectural direction can be communicated and governed. These artefacts can support consistency by giving organisational stakeholders a shared point of reference when planning initiatives or designing solutions.

The relationship between TOGAF and organisational practice nevertheless requires careful interpretation. Kotusev's (2018) case study found that several TOGAF elements were either partially adopted or not adopted, while adapted reference models and organisation-specific governance arrangements were used instead. The significance of a framework therefore lies not only in whether it is formally adopted, but in how its concepts and artefacts become integrated into organisational decision-making.

In the context of this thesis, TOGAF is relevant because it establishes the importance of governing and maintaining architectural knowledge. However, the presence of repositories and formal artefacts does not resolve how stakeholders identify the appropriate material or interpret it within a particular initiative. This limitation motivates the study's attention to EA as a knowledge-access and guidance practice.

2.1.2 Enterprise Architecture as Organisational Alignment

A central purpose of EA is to support alignment between organisational direction and technology-related decisions. Ross, Weill and Robertson (2006) argue that organisations create an execution foundation by establishing appropriate levels of process integration and standardisation. EA represents the organising logic through which this foundation is expressed and developed.

The alignment perspective extends beyond the technical compatibility of systems. Decisions concerning applications, information, platforms, integrations, capabilities, and governance can affect several organisational areas. EA provides concepts and representations through which these dependencies can be considered before local decisions generate wider inconsistency.

This perspective also explains why EA knowledge has an audience beyond Enterprise Architects. Solution architects, product owners, business stakeholders, engineers, and governance functions may need architectural information when planning initiatives or making implementation decisions. The practical influence of EA therefore depends on whether these stakeholders can recognise the relevant architectural direction and apply it to their circumstances.

The internal description of the case organisation's EA capability reflects this relationship. The function is described as structuring business, information, and technology landscapes, reducing complexity, designing target architectures, shaping transition plans, and guiding solution architects and product owners. These activities depend on communication and stakeholder engagement in addition to the production of architecture artefacts.

Accordingly, EA effectiveness should not be assessed solely by the completeness of its models. Its contribution also depends on whether architectural knowledge becomes usable in practice. This shifts the literature review towards knowledge management and the conditions under which documented and experience-based knowledge can be shared across organisational boundaries.

2.2 Knowledge Management in Enterprise Architecture

Knowledge management concerns the processes through which organisational knowledge is created, formalised, shared, maintained, and reused. Davenport and Prusak (1998) describe organisational knowledge as a combination of experience, values, contextual information, and expert insight that enables the evaluation and incorporation of new information. This understanding distinguishes knowledge from information stored in isolation because knowledge acquires practical value through interpretation and use.

The relevance of knowledge management to EA follows from the knowledge-intensive character of architecture practice. Enterprise Architects produce and interpret knowledge concerning capabilities, processes, applications, information, technologies, integrations, standards, governance, and organisational change. This knowledge is represented through formal artefacts but is also developed through discussion, professional judgement, and accumulated experience.

Nonaka and Takeuchi (1995) distinguish between explicit and tacit knowledge. Explicit knowledge can be formalised and communicated through documents, procedures, models, databases, and other representational forms. In EA, architecture principles, guardrails, capability maps, target architectures, roadmaps, process descriptions, and decision records constitute explicit knowledge.

Tacit knowledge is connected to experience, judgement, intuition, and contextual understanding. It is more difficult to formalise because practitioners may understand how to respond to a situation without being able to represent every element of that understanding in a document. Within EA, tacit knowledge can include awareness of earlier decisions, organisational relationships, practical exceptions, stakeholder concerns, and the reasoning through which general guidance is applied to a particular initiative.

Nonaka and von Krogh (2009) further explain that tacit and explicit knowledge should not be understood as separate categories. Knowledge can move along a continuum as elements of experience are articulated and formalised. This is relevant to EA because architectural discussions may result in new principles, decisions, examples, or guidance, while documented knowledge becomes meaningful when stakeholders interpret and apply it.

Nonaka and Takeuchi's (1995) SECI model describes knowledge creation through four modes of conversion: socialisation, externalisation, combination, and internalisation. Socialisation concerns the sharing of tacit knowledge through interaction and experience. Externalisation converts aspects of tacit knowledge into explicit representations. Combination integrates existing explicit knowledge into new structures, while internalisation occurs when individuals develop understanding using explicit knowledge.

These processes are visible in EA practice. Interaction between architects and stakeholders enables contextual knowledge to be shared. Principles, models, and decisions formalise parts of this understanding. Repositories and architecture models combine information from several sources, while stakeholders internalise architectural knowledge by applying it in initiatives and designs.

The difficulty identified in the literature is that codification alone does not ensure effective knowledge use. A repository may contain extensive documentation while stakeholders remain uncertain about what to search for, which source is authoritative, or how general guidance relates to a specific situation. Davenport and Prusak (1998) emphasise that knowledge management involves people and organisational processes rather than storage technology alone.

Toumi, Fosso Wamba and Hafsi (2025) bring this argument directly into the EA domain. Their review positions EA as a knowledge-management discipline and identifies fragmented silos, loss of tacit knowledge, semantic underspecification, tool limitations, and governance misalignment as barriers to organisational learning and strategic alignment. The authors therefore support the argument that architecture repositories and governance artefacts must be considered in relation to the organisational processes through which knowledge is interpreted and reused.

For the present study, knowledge management explains why improving EA support cannot be reduced to connecting more documents to a search interface. The proposed artefact must provide access to explicit knowledge while acknowledging that some questions still require context, professional judgement, or escalation to an expert. The distinction between explicit and tacit knowledge therefore informs both the research problem and the human-oversight requirements of the artefact.

2.3 AI-Supported Knowledge Retrieval

The growth of organisational information has increased the importance of technologies that help employees identify relevant content across internal repositories. Information retrieval traditionally concerns the representation, storage, organisation, and discovery of information in response to a user query (Manning, Raghavan and Schütze 2008). Within an enterprise setting, search must also account for organisational terminology, content quality, metadata, user requirements, and the management of the search environment (White 2015). Enterprise search should therefore be understood as an organisational capability rather than solely as a search interface.

Developments in Large Language Models have introduced new ways of interacting with retrieved information. LLMs can generate natural-language responses from supplied context, but their output may contain unsupported or factually inconsistent statements (Brown et al.

2020; Ji et al. 2023). Retrieval-Augmented Generation addresses part of this limitation by combining a generative model with access to an external information source. In the original RAG formulation, retrieved passages provide non-parametric memory that supplements the knowledge represented in the model's parameters (Lewis et al. 2020).

Recent reviews show that RAG has developed into a wider set of retrieval and generation techniques involving dense, sparse, hybrid, and structure-aware retrieval. However, their effectiveness differs across tasks and evaluation practices, and retrieval errors can still contribute to unsupported responses (Brown, Roman and Devereux 2025; Zhang and Zhang 2025). RAG should therefore not be presented as eliminating hallucination. Its value lies in providing retrieved evidence that can improve grounding and make the informational basis of a response more visible.

For enterprise knowledge management, the main significance of RAG is its ability to connect conversational interaction with organisation-specific information. Karakurt and Akbulut's (2026) systematic review nevertheless finds that enterprise adoption remains largely experimental and identifies a gap between academic prototypes and integration into production workflows. This supports the need to examine organisational readiness, source quality, governance, and evaluation alongside technical retrieval performance.

2.3.1 Enterprise Search and Knowledge Discovery

Enterprise search refers to the retrieval of information across internal organisational sources such as document repositories, intranets, databases, and collaboration platforms. Traditional information retrieval methods use document representations, query representations, indexes, and ranking mechanisms to identify material relevant to a user's information need (Manning, Raghavan and Schütze 2008).

Enterprise search presents additional organisational considerations. Content may be distributed across systems, described through inconsistent terminology, or maintained under different ownership arrangements. White (2015) argues that an effective enterprise search environment requires attention to user requirements, content quality, metadata, evaluation, and continuing organisational support. Search quality consequently depends on both the retrieval technology and the condition of the information environment.

This distinction is relevant to EA because architectural knowledge may be represented through principles, guardrails, capability models, roadmaps, process descriptions, and decision records. The retrieval problem is not limited to identifying documents containing matching terms. Users must also determine whether the information is authoritative and relevant to the decision or process under consideration.

2.3.2 Retrieval-Augmented Generation

Retrieval-Augmented Generation combines language generation with information retrieved from an external source. Lewis et al. (2020) introduced RAG as an approach that combines the parametric memory of a language model with non-parametric memory represented through a retrievable document index. The retrieved passages are supplied as context during response generation, allowing the model to draw upon information outside its trained parameters.

A typical RAG process involves representing source content for retrieval, identifying passages relevant to a question, and supplying selected passages to the language model. The generated response is therefore conditioned on both the question and the retrieved content. This can support domain-specific and updateable knowledge use without requiring the complete knowledge base to be incorporated into the model itself (Lewis et al. 2020).

However, grounding a model through retrieval does not guarantee factual correctness. Retrieval may return irrelevant, incomplete, or conflicting passages, while the generation stage may interpret otherwise suitable evidence incorrectly. Reviews of RAG evaluation and hallucination mitigation therefore emphasise the need to assess the retrieval and generation components together rather than assuming that retrieval automatically produces trustworthy answers (Brown, Roman and Devereux 2025; Zhang and Zhang 2025).

This limitation is significant for EA guidance because the underlying documentation may itself be fragmented, outdated, or incomplete. RAG can improve access to available explicit knowledge, but it cannot independently resolve weaknesses in the sources from which that knowledge is retrieved.

2.3.3 Knowledge Assistants and Conversational Support Systems

Knowledge assistants combine conversational interaction with access to a defined information environment. Unlike conventional search interfaces, which primarily return ranked results, conversational systems can formulate responses and allow users to refine their information needs through follow-up questions.

Research on conversational agents identifies natural-language processing, communication, trust, user adoption, and organisational value as central areas of investigation (Mariani, Hashemi and Wirtz 2023). These findings indicate that the usefulness of a conversational assistant depends not only on response generation but also on whether users regard the interaction as relevant and trustworthy.

In the context of EA, a knowledge assistant could support document discovery and explanation of established concepts. However, its responses remain dependent on the knowledge sources, retrieval configuration, and boundaries within which it operates. Questions requiring

exceptions, professional judgement, or formal decisions may remain unsuitable for automated resolution.

2.3.4 Enterprise Copilots and AI-Supported Workplace Assistance

Enterprise copilots represent conversational assistants integrated into workplace environments and organisational information sources. Conceptually, they combine natural-language interaction with retrieval, content generation, and, in some implementations, workflow support.

For this thesis, enterprise copilots are relevant because they provide a possible interface between employees and existing EA knowledge. Their organisational value, however, cannot be inferred from technical integration alone. Trust, source transparency, access control, maintenance, and human oversight remain necessary where generated responses may influence organisational action. These considerations provide the transition from AI-supported retrieval to the broader discussion of agents and enterprise assistants in the following section.

2.4 AI Agents and Enterprise Assistants

AI agents extend conversational interaction by supporting goal-oriented activities that may involve planning, memory, information retrieval, tool use, and action. Wang et al. (2024) present a framework for understanding LLM-based autonomous agents and review their construction, applications, and evaluation. The review also identifies continuing challenges and future research directions, indicating that agentic capability should not be treated as equivalent to dependable organisational autonomy.

In this study, it is useful to distinguish among four related concepts. A conversational agent enables dialogue through natural language. A knowledge assistant connects that interaction to a bounded information domain. An enterprise copilot embeds assistance within an organisational work environment. An agentic system may additionally use tools or coordinate multiple steps towards a goal. These categories can overlap, but they should not be treated as interchangeable.

2.4.1 Conversational Agents

Conversational agents are systems designed to interact with users through language-based dialogue. Research concerning these systems encompasses natural-language processing, communication quality, trust, adoption, and value creation for users and organisations (Mariani, Hashemi and Wirtz 2023).

Within the present study, conversational interaction provides an alternative to navigating repositories and folder structures. Its intended contribution is to make it easier for stakeholders to express an information need. The conversational interface does not itself ensure that the answer is correct. Response quality remains dependent on retrieval, source quality, system instructions, and the boundaries of the connected knowledge environment.

2.4.2 Agentic Systems

Agentic systems extend question answering by supporting activities directed towards an objective. LLM-based agents may incorporate components related to planning, memory, environmental interaction, and action. Wang et al. (2024) review these components and the evaluation strategies used across agent applications.

For EA support, agentic behaviour may be relevant when a user must be guided through a process rather than given a single answer. A system may collect required information, retrieve relevant guidance, request clarification, and route the user to an appropriate expert. Nevertheless, each additional action increases the importance of access control, monitoring, accountability, and human oversight.

2.4.3 Knowledge Assistants

Knowledge assistants are conversational systems connected to selected information sources within a bounded domain. Their intended function is to help users locate or understand knowledge relevant to a task. In an EA setting, the domain may include architecture principles, guardrails, capability information, process guidance, and ownership information.

The bounded nature of the assistant is significant. Connecting the assistant to approved sources allows its intended knowledge scope to be defined, although this does not guarantee that every generated statement will be supported. Source references, evaluation, escalation, and continuing content maintenance are therefore required.

2.4.4 Enterprise Copilots

Enterprise copilots integrate conversational assistance into workplace platforms and organisational information environments. They may combine retrieval, response generation, process guidance, and access to organisational tools. In the present study, the enterprise-copilot concept provides the organisational interface through which stakeholders interact with the proposed EA guidance artefact.

The concept is based on augmentation rather than the autonomous replacement of architectural expertise. This distinction is important because EA questions may range from low-risk document discovery to high-risk interpretation of standards or governance exceptions. The suitability of an enterprise copilot therefore depends on the type of question, the authority of its sources, and the availability of human review.

Taken together, the literature on conversational agents, knowledge assistants, enterprise copilots, and agentic systems provides a basis for defining the capabilities and boundaries of the proposed artefact. It also reinforces the need to evaluate the assistant as part of an organisational knowledge and governance environment rather than as an isolated language model.

2.5 Design Science Research

Design Science Research (DSR) is a problem-solving research paradigm concerned with the creation and evaluation of artefacts intended to extend human or organisational capabilities. Within Information Systems research, DSR is appropriate when the research objective involves not only understanding an organisational problem but also designing and evaluating a purposeful response to it. Hevner et al. (2004) position DSR at the intersection of people, organisations, and technology and argue that knowledge of a problem and its possible solution is developed through the construction and application of an artefact.

Artefacts in Information Systems research may include constructs, models, methods, or implemented systems. The academic contribution of a DSR study does not therefore depend solely on producing a fully operational technical product. It may also arise from a justified design, a method for addressing a class of organisational problems, or knowledge derived from the development and evaluation process. Gregor and Hevner (2013) emphasise that DSR contributions must be positioned in relation to existing knowledge in both the problem and solution domains.

The present study follows the Design Science Research Methodology proposed by Peffers et al. (2007). The methodology comprises six activities: problem identification and motivation, definition of objectives for a solution, design and development, demonstration, evaluation, and communication. The process provides a structured connection between an identified organisational problem and the development of an artefact intended to address it.

The suitability of DSR for this study follows from the nature of the research problem. The thesis does not examine EA knowledge access solely to describe existing organisational conditions. It also derives design requirements, evaluates alternative AI approaches, develops a prototype, and assesses that prototype within a selected pilot domain. DSR therefore provides the

methodological logic connecting the empirical investigation with artefact development and evaluation.

In this study, interviews, document analysis, and workshop supported problem identification and the definition of solution objectives. The validated stakeholder needs were translated into design requirements, which informed the comparison of alternative AI approaches and the development of the prototype. The prototype was subsequently demonstrated and evaluated in relation to representative EA use cases and expert assessment. The thesis itself represents the communication stage of the DSR process.

This application of DSR is consistent with Gregor and Hevner's (2013) argument that a DSR contribution should include not only the artefact but also the knowledge generated through its design and evaluation. In the present study, this knowledge concerns the relationship between EA knowledge-access challenges, source readiness, stakeholder requirements, governance, architecture selection, and AI-supported guidance.

2.6 Analytical Framework

The literature reviewed in this chapter identifies a connection between Enterprise Architecture practice, knowledge management, and AI-supported retrieval. The analytical framework shown in Figure 2.1 synthesises these perspectives and establishes the conceptual logic used to interpret the empirical findings and guide artefact development.

Enterprise Architecture produces and maintains knowledge through principles, guardrails, capability models, process descriptions, roadmaps, governance arrangements, and decision-related artefacts. However, EA should not be understood solely through these formal outputs. Its organisational value also depends on how architectural knowledge is communicated and used by stakeholders involved in business and technology decisions (Ross, Weill and Robertson 2006; Lankhorst 2017; Kotusev 2018).

Knowledge-management theory explains why the existence of formal artefacts does not necessarily ensure effective knowledge use. Explicit knowledge can be represented in documents, models, and repositories, whereas contextual understanding and professional judgement may remain tacit and closely associated with experienced practitioners (Nonaka and Takeuchi 1995; Nonaka and von Krogh 2009). Where formal knowledge is fragmented, unclear, or difficult to interpret, stakeholders may continue to depend on individual experts.

AI-supported retrieval represents a possible mechanism for improving access to explicit organisational knowledge. Enterprise search can help users discover relevant content, while RAG can connect retrieved evidence with natural-language response generation (Manning, Raghavan and Schütze 2008; Lewis et al. 2020). However, retrieval does not independently

guarantee accuracy. Response quality remains dependent on source quality, retrieval performance, and the way retrieved evidence is used during generation (Brown, Roman and Devereux 2025; Zhang and Zhang 2025).

The framework therefore places design requirements between AI-supported retrieval and artefact development. These requirements represent the organisational conditions that must shape the technical solution, including source authority, contextual explanation, ownership, access control, transparency, escalation, and human oversight.

Taken together, the literature suggests a logical progression from the existence of Enterprise Architecture knowledge to the design of an AI-supported guidance artefact. This progression forms the analytical framework for the study and illustrates how the various theoretical perspectives contribute to understanding the research problem and informing the design process.

As illustrated in Figure 2.1, the framework begins with EA knowledge and the organisational practices through which this knowledge is produced and used. Knowledge-access challenges motivate the investigation of AI-supported retrieval, but the technical possibilities are filtered through stakeholder needs and organisational constraints before being translated into artefact requirements. The framework is used in Chapter 5 to interpret the empirical findings and derive design requirements. Chapter 6 applies these requirements when comparing alternative AI architectures and developing the prototype. Chapter 7 evaluates the resulting artefact against the requirements established through this process.

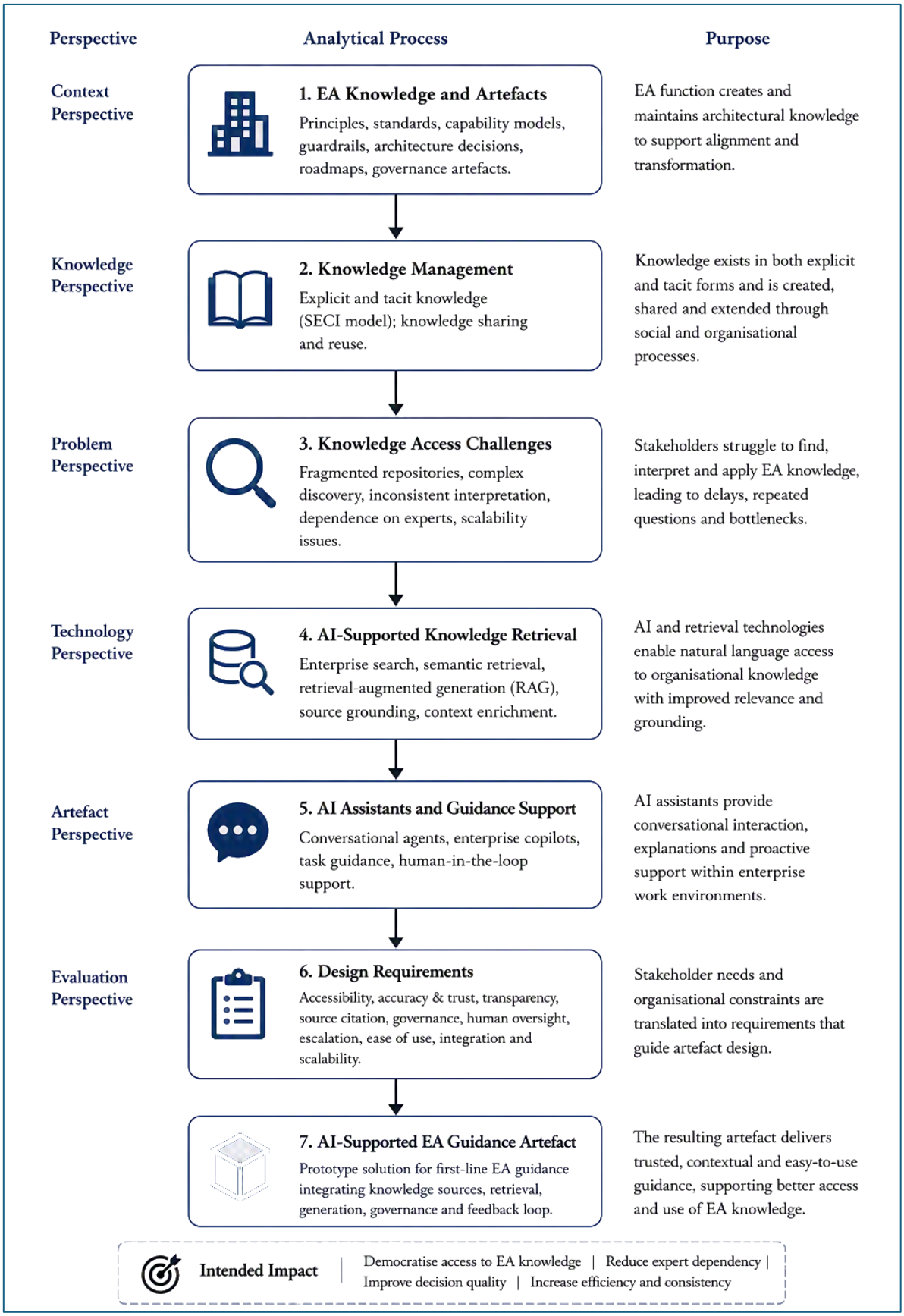


Figure 2.1. Analytical Framework Guiding the Study

2.7 Literature Gap and Research Positioning

The reviewed literature provides established explanations of Enterprise Architecture, organisational knowledge, information retrieval, conversational agents, and Design Science Research. However, the intersection of these fields remains less developed.

EA literature primarily explains how architecture supports organisational alignment, governance, and coordinated change through principles, models, methods, and architecture practices (Ross, Weill and Robertson 2006; Lankhorst 2017; The Open Group 2022). Practice-oriented research further shows that organisations adapt formal EA frameworks and combine architecture artefacts with organisation-specific governance arrangements and terminology (Kotusev 2018). These contributions explain the function and organisation of EA but provide less direct guidance concerning how diverse stakeholders can interact with architectural knowledge through AI-supported guidance.

Knowledge-management literature explains how explicit and tacit knowledge are created, formalised, shared, and reused (Davenport and Prusak 1998; Nonaka and Takeuchi 1995; Nonaka and von Krogh 2009). These perspectives help explain why EA documentation cannot fully capture the contextual understanding held by experienced architects. However, this literature does not by itself specify how AI-supported artefacts should combine formal sources, contextual explanation, ownership, and expert escalation within EA practice.

Toumi, Fosso Wamba and Hafsi (2025) provide more direct support for this intersection by positioning EA as a knowledge-management discipline. Their review identifies knowledge fragmentation, tacit-knowledge loss, semantic under specification, tool constraints, and governance misalignment as barriers and proposes a research agenda for AI-augmented EA systems. This recent contribution indicates that AI-supported EA is an emerging research direction and that further empirical and design-oriented investigation is required.

RAG and conversational-agent research provide relevant technical foundations. RAG enables generative models to draw upon retrieved external information, while conversational agents provide natural-language interaction with users (Lewis et al. 2020; Mariani, Hashemi and Wirtz 2023). Nevertheless, reviews of RAG research identify continuing variation in techniques, evaluation practices, and empirical performance (Brown, Roman and Devereux 2025). Research focused specifically on enterprise knowledge management also reports that many RAG implementations remain experimental and that comparatively few studies address the requirements of real-time production integration (Karakurt and Akbulut 2026).

These findings qualify the claim that AI-supported retrieval is already an established solution to enterprise knowledge-access problems. The literature demonstrates its technical potential, but it also shows that retrieval quality, source preparation, evaluation, organisational integration, and governance remain important areas of investigation.

A further gap concerns the translation of organisational evidence into artefact design. Existing research explains individual elements of the problem, including EA practice, organisational knowledge, retrieval technology, and conversational interaction. However, less is known about how stakeholder challenges, source readiness, organisational constraints, and alternative AI architectures can be combined within a systematic design process for EA guidance.

The present study is positioned within this gap. It examines EA knowledge access as an organisational problem, derives artefact requirements from stakeholder evidence, evaluates alternative AI approaches, and develops and evaluates a bounded guidance artefact through DSR. The intended contribution is therefore not a new language model or retrieval algorithm. It is a stakeholder-informed design and evaluation of AI-supported EA guidance within an industrial case.

This positioning connects the study to two bodies of knowledge. First, it contributes to EA and knowledge-management research by examining how architectural knowledge may become more accessible beyond the specialist architecture community. Second, it contributes to applied enterprise-AI research by showing how knowledge readiness, ownership, governance, architecture selection, and human oversight influence the design of a domain-specific guidance artefact.

3. Research methodology

3.1 Introduction to the Research Methodology

This chapter presents the methodological approach adopted to address the research problem and achieve the objectives of the study. As established in the previous chapter, the study seeks not only to understand challenges related to Enterprise Architecture (EA) knowledge accessibility but also to design a purposeful artefact capable of addressing those challenges within an industrial context. Consequently, the research requires a methodology that supports both investigation and design.

Traditional explanatory or descriptive research approaches are often primarily concerned with understanding phenomena. While such approaches can provide valuable insight into organisational challenges, they are less suited to situations where the objective is to create and evaluate a practical solution. Since the present study aims to develop an AI-supported guidance artefact informed by stakeholder needs and organisational requirements, a design-oriented research approach is required.

For this reason, the study adopts Design Science Research (DSR) as its overarching methodological foundation. Design Science Research is widely used within Information Systems research when the objective is to address a real-world problem through the creation and evaluation of an artefact. Rather than treating understanding and solution development as separate activities, DSR combines both perspectives by linking empirical investigation with purposeful design.

The methodological process adopted in this study follows the Design Science Research Methodology (DSRM) proposed by Peffers et al. (2007). The methodology provides a structured process consisting of problem identification and motivation, definition of solution objectives, design and development, demonstration, evaluation, and communication. These activities provide a systematic framework for moving from an observed organisational challenge towards a justified artefact design while maintaining a clear connection between empirical evidence and design decisions.

The chapter is structured around the major activities of the DSR process. It first describes the overall research philosophy and research design before explaining the case organisation, data collection procedures, and analytical approach used to investigate the problem context. Subsequently, the chapter explains how empirical findings were translated into design requirements, how candidate solution approaches were evaluated, and how the proposed artefact was developed and assessed. Finally, the chapter addresses research quality considerations and ethical aspects relevant to the study.

3.2 Design Science Research Methodology

Design Science Research is a problem-solving research paradigm concerned with the creation and evaluation of innovative artefacts intended to address identified organisational problems. Within Information Systems research, DSR differs from purely explanatory research traditions because its primary objective is not only to describe or predict phenomena but also to develop purposeful solutions capable of improving existing conditions.

Hevner et al. (2004) argue that Design Science Research contributes knowledge through the design, construction, and evaluation of artefacts such as models, methods, frameworks, systems, and processes. These artefacts are developed to solve relevant organisational problems while simultaneously generating knowledge that extends beyond the immediate application context. As a result, DSR occupies a unique position between academic inquiry and practical problem-solving.

The suitability of Design Science Research for the present study arises from the nature of the research problem itself. The challenge identified within Sandvik Coromant is not merely a lack of understanding regarding Enterprise Architecture knowledge accessibility. Rather, the organisation seeks a practical mechanism capable of improving how such knowledge can be discovered, interpreted, and reused by stakeholders. Addressing this problem therefore requires both investigation of the current situation and the development of a purposeful artefact capable of providing support.

Among the various Design Science approaches available, this study adopts the Design Science Research Methodology (DSRM) proposed by Peffers et al. (2007). The methodology is particularly appropriate because it provides a structured sequence of activities that connect problem understanding with artefact creation and evaluation. The six activities of DSRM are:

1. Problem Identification and Motivation
2. Definition of Objectives for a Solution
3. Design and Development
4. Demonstration
5. Evaluation
6. Communication

Rather than treating these activities as strictly sequential, DSRM recognises that iterations may occur throughout the research process as new insights emerge and design decisions are refined. This flexibility is particularly valuable in organisational settings where requirements and constraints often become clearer through stakeholder interaction and prototype development.

The first activity, problem identification and motivation, focused on understanding existing challenges associated with Enterprise Architecture support and knowledge accessibility. This phase relied on stakeholder interviews, workshop activities, and document analysis to identify recurring problems, expectations, and opportunities for improvement.

The second activity involved defining objectives for a solution. Insights generated during the empirical investigation were translated into design requirements describing the capabilities, characteristics, and constraints that an AI-supported guidance artefact should satisfy. These requirements formed the foundation for subsequent design activities.

The design and development phase concerned the exploration and assessment of alternative AI approaches, including enterprise copilots, retrieval-augmented generation architectures, and other AI-supported retrieval mechanisms. Design decisions were guided by stakeholder requirements, organisational considerations, governance constraints, and technological feasibility.

Demonstration focused on translating the proposed solution into a prototype-oriented artefact concept capable of illustrating how AI-supported guidance could operate within the selected pilot domain. Evaluation subsequently examined the extent to which the proposed artefact addressed the identified problem and satisfied design requirements derived from empirical findings.

Finally, communication is represented by the thesis itself, which documents the research process, artefact development activities, evaluation outcomes, and resulting contributions to both practice and research.

The use of DSRM provides methodological coherence throughout the study by ensuring that all design decisions remain traceable to the organisational problem and stakeholder needs identified during the empirical investigation. This traceability is particularly important because the objective of the study is not simply to implement a technology but to develop an artefact that is both organisationally relevant and theoretically informed.

The six DSRM activities provided the methodological logic of the study. For project planning and reporting, these activities were operationalised through the seven research stages shown in Table 3.1. The four phases presented in the abstract provide a higher-level summary of the same process rather than a separate methodology.

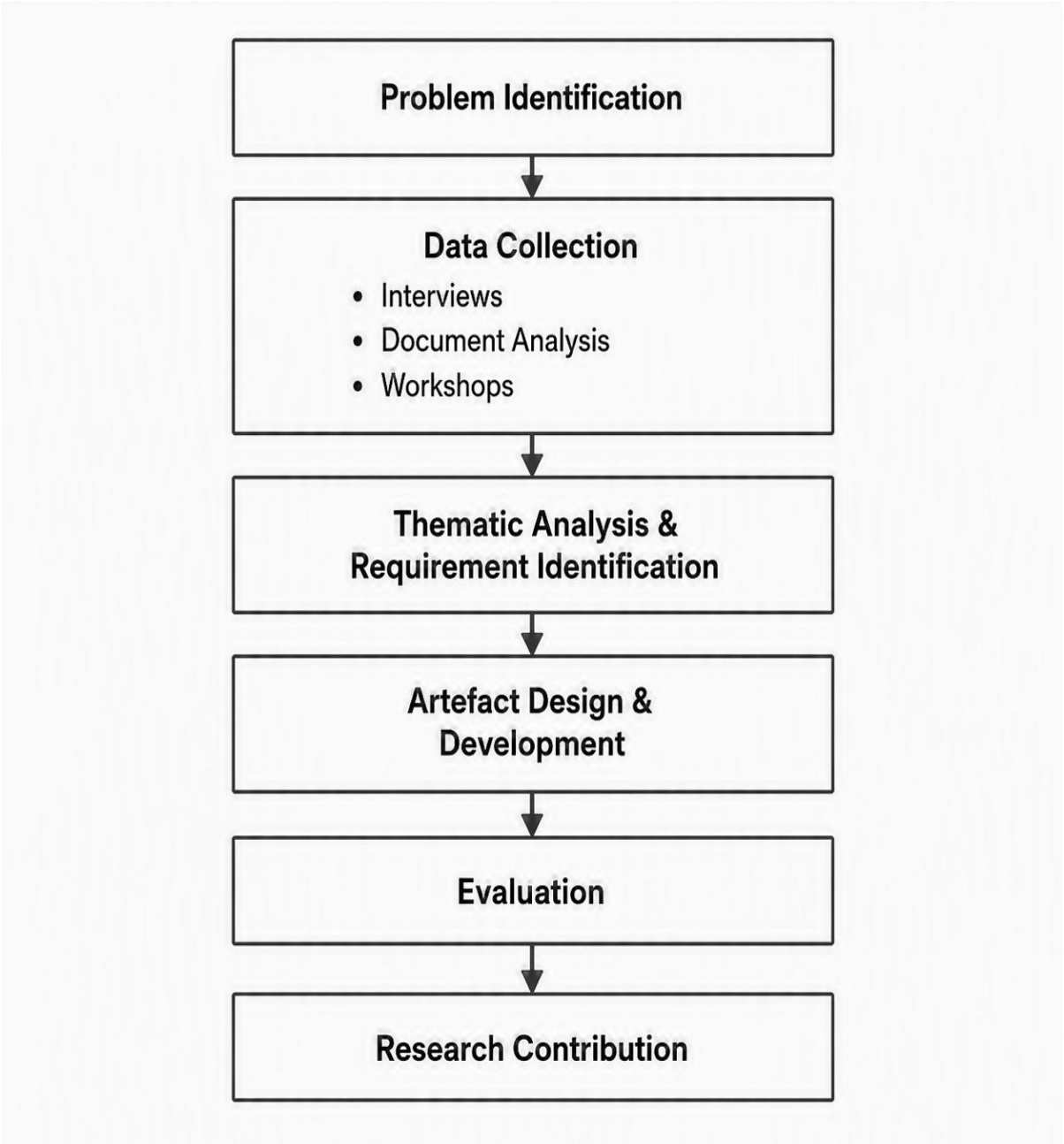


Figure 3.1. Research Process Based on Design Science Research Methodology

3.3 Case Organisation

The empirical setting of the study is the Enterprise Architecture environment at Sandvik Coromant. The case is situated within a global industrial Organisation where digital transformation, governance, standards, and cross-functional coordination are important for how technology-related decisions are made. Public information describes Sandvik as a global engineering group operating in more than 150 countries, while Sandvik Coromant is positioned within this broader context as a provider of machining solutions, tools, technical knowledge, and digital manufacturing support. This makes the case relevant for studying how architecture knowledge is created, distributed, and used in practice.

A single-case study was chosen because the research aims to achieve a detailed understanding of a contemporary phenomenon in its real-world context. Case study research is particularly appropriate when the research addresses how questions, when the phenomenon is context dependent, and when multiple sources of evidence are needed. Following Yin (2018), the case study integrated interviews, internal documents, workshop material, and artefact-related design activities within a single organisational setting.

3.4 Overview of the Research Process

The study was conducted iteratively in accordance with the Design Science Research Methodology. The research progressed from problem investigation and stakeholder engagement to use-case validation, architecture evaluation, prototype development, and evaluation. Table 3.1 summarises the principal research activities and their outputs.

The period descriptions in Table 3.1 represent the principal stage of each activity. The process was iterative, and some activities overlapped. For example, document review continued during prototype development, while stakeholder feedback informed both the architecture comparison and refinement of the artefact. The workshop took place on 19 May 2026 and translated the preliminary use cases into representative questions, ownership responsibilities, knowledge-source mappings, and MVP candidates. Technical investigation and prototype work continued during July and August, including preparation of SharePoint material for retrieval, exploration of vector and graph-based approaches, and retention of Copilot Studio as the principal MVP platform.

The research process was not entirely linear. Insights obtained during technical experimentation occasionally revealed weaknesses in source material or retrieval readiness, requiring the study to return to document preparation and knowledge-source assessment. This iterative movement between problem understanding, design, and evaluation is consistent with the design-oriented character of the study.

Table 3.1. Overview of Research Activities and Outputs

Research stage	Period	Principal activities	Main outputs
Problem investigation	March to April 2026	Semi-structured stakeholder interviews and initial review of EA documents	Identified knowledge-access challenges, recurring questions, and potential AI-support use cases
Use-case consolidation	April to May 2026	Open coding, consolidation of related observations, and organisation of use cases into preliminary clusters	Preliminary use cases and nine higher-level use-case clusters
Workshop validation	19 May 2026	Use-case validation, question formulation, task classification, ownership assignment, document mapping, and MVP prioritisation	Validated use cases, representative question bank, ownership map, source-readiness observations, and candidate pilot areas
Solution investigation	June to July 2026	Comparison of Copilot Studio, custom RAG, graph-oriented retrieval, and open-source approaches	Architecture comparison and evaluation criteria
Knowledge preparation and prototype development	June to August 2026	Curation of SharePoint sources, removal of unsuitable content, configuration of knowledge retrieval, and development of controlled Start Assurance workflows	Curated knowledge repository and Copilot Studio-based prototype
Prototype demonstration and formative evaluation	July to August 2026	Demonstration of workflows, review of generated responses, collection of stakeholder feedback, and identification of human-review requirements	Evaluation observations, strengths, limitations, governance requirements, and improvement areas
Synthesis and communication	August 2026	Integration of empirical findings, design rationale, evaluation findings, and research contributions	Thesis report, proposed MVP roadmap, and scaling recommendations

3.5 Data Collection

The data collection process was designed to support the objectives of the study by generating a detailed understanding of the challenges associated with Enterprise Architecture (EA) knowledge accessibility, stakeholder requirements, and the organisational context within which an AI-supported artefact would be developed. Consistent with Design Science Research (DSR), data collection served both exploratory and design-oriented purposes. The activities were therefore structured to support problem identification, requirements elicitation, artefact development, and evaluation throughout the research process.

To achieve methodological triangulation and strengthen the credibility of the findings, data were collected using multiple sources of evidence. Semi-structured interviews were conducted to capture stakeholder perspectives and identify existing challenges related to EA knowledge access. Document analysis was used to examine organisational artefacts, architecture-related documentation, and existing knowledge repositories. In addition, collaborative workshop was organised with key stakeholders to validate findings, identify high-value use cases, prioritise requirements, and guide artefact design decisions.

The combination of these data collection methods enabled the study to capture both individual experiences and collective organisational perspectives, thereby providing a robust foundation for the subsequent analysis and design activities.

3.5.1 Interviews

Semi-structured interviews constituted the primary source of empirical evidence during the problem-identification stage of the study. The interviews were conducted to understand how stakeholders currently access and use Enterprise Architecture knowledge, the difficulties encountered in this process, and the areas in which AI-supported guidance could provide practical value.

Participants were selected purposively based on their relationship to the creation, governance, interpretation, or use of EA knowledge. The sample included architecture roles and stakeholders from product, operations, automation, security, and AI-related domains. The purpose was not to achieve statistical representation of the wider organisation, but to obtain perspectives from participants with different forms of involvement in EA practice.

A total of 12 semi-structured interviews were completed and included in the formal analysis. The meeting tracker distinguished these completed interviews from contacts that remained pending, tentative, unanswered, or unscheduled. Only interviews explicitly recorded as completed were treated as formal interview data.

To preserve participant confidentiality, the interviewees are presented through broad role categories rather than by name, exact job title, or organisational unit. Table 3.2 provides an anonymised overview of the interview sample.

Table 3.2. Anonymised interview participant groups, relationships to EA knowledge, and contributions.

Participant group	Total Participants	Relationship to EA knowledge	Principal perspective contributed
Enterprise and Business architecture	5	Create, maintain, interpret, or govern architectural knowledge	EA processes, business capabilities, guardrails, knowledge structures, and organisational alignment
Solution architecture	3	Applies EA guidance during solution design	Knowledge retrieval, fragmented information, source transparency, and practical guidance
Product Ownership & Operations	2	Use or enable EA-related knowledge in operational settings	Architectural awareness, repository maintenance, automation, and organisational adoption
Digital security	1	Contributes specialist knowledge concerning security and risk	Security constraints, risk-sensitive use cases, and human-review requirements
AI platform & Automation	1	Assesses AI platform and data-related considerations	Platform suitability, organisational data use, feedback, and technical integration
Total	**12**		

The categories in Table 3.2 are intentionally broad. Where a participant's responsibilities crossed several domains, the participant was assigned to the category that best represented the principal perspective contributed during the interview. This approach avoided double counting while reducing the possibility of identifying individual participants. The interviews followed a common semi-structured guide covering the following areas:

- current access to and use of EA knowledge;
- recurring EA questions and support needs;
- difficulties locating or interpreting architectural guidance;

- reliance on architects or other subject-matter experts;
- potential applications of AI-supported EA guidance;
- expectations concerning source transparency, trust, escalation, and human review.

The semi-structured format ensured consistent coverage of the study topics while allowing participants to elaborate on experiences relevant to their respective roles. The completed interviews generated evidence concerning the organisation of EA documentation, fragmentation of knowledge across systems, retrieval and summarisation needs, interpretation of EA principles, repository maintenance, security constraints, integration-related knowledge, capability information, and AI-platform considerations.

During the analysis, recurring problems, questions, and desired forms of support were assigned descriptive codes. Related observations were subsequently consolidated into preliminary AI-support use cases and organised into higher-level use-case clusters.

Additional stakeholder discussions and project-related meetings contributed to the broader exploration of possible use cases. However, these discussions were not treated as completed research interviews. This distinction is important because the use-case summary contains contributions attributed to individuals who were not recorded as completed interview participants in the meeting tracker. The additional material informed the exploration and refinement of possible use cases, while the formal empirical interview dataset remained limited to the 12 confirmed interviews.

The preliminary use-case clusters were subsequently introduced into the workshop for collective review and refinement. The workshop examined their relevance, scope, ownership, required knowledge sources, and suitability for prototype development. The clusters were therefore treated as intermediate analytical outputs rather than direct or final representations of individual interview statements.

The interview sample should not be regarded as statistically representative of Sandvik Coromant as a whole. Instead, it provided variation across roles with different relationships to EA knowledge. This enabled the study to consider perspectives from stakeholders who create and govern architectural knowledge alongside those who consume, apply, or technically enable it.

3.5.2 Document Analysis

Document analysis was conducted to complement the interviews and develop a clearer understanding of the formal Enterprise Architecture knowledge available within the case organisation. While the interviews captured stakeholder experiences and knowledge-access

challenges, the document analysis examined the organisational artefacts through which EA guidance was documented, maintained, and communicated.

The documents were selected purposively based on their relevance to the EA support needs and potential use cases identified during the empirical investigation. The selection was therefore not intended to represent every document maintained by the EA function. Instead, attention was directed towards artefacts that supported recurring stakeholder questions, the selected pilot domain, or the assessment of potential knowledge sources for the proposed artefact.

The reviewed material included the following document categories:

- Enterprise Architecture guardrails and principles,
- Start Assurance process descriptions, self-assessment material, meeting templates, and supporting guidance;
- Business capability information and capability ownership data;
- Governance and engagement-related material;
- Architectural guidance and process descriptions;
- Selected internal presentations and SharePoint pages relevant to the EA function.

The document analysis served three purposes. First, the material provided contextual understanding of the EA function, its terminology, and its established guidance processes. Second, it enabled comparison between stakeholder descriptions and the knowledge formally available within organisational repositories. Third, it supported an assessment of whether identified questions could be answered using existing and authoritative sources.

The analysis considered each document's relevance, apparent currency, duplication, ownership, accessibility, and suitability for AI-supported retrieval. Particular attention was given to whether the material contained sufficient information to support the intended question and whether the authoritative source could be identified. The workshop subsequently formalised this assessment by mapping representative questions to required documents and considering the readiness of the available knowledge. Workshop discussions confirmed that Start Assurance process steps were available in SharePoint and were reviewed for currency. They also identified desirable but unavailable material, including a decision log documenting previous exceptions to guardrails.

The document analysis demonstrated that the availability of information did not necessarily indicate readiness for AI-supported use. Some knowledge was distributed across several artefacts, while other material required clarification, consolidation, or explicit ownership.

Workshop participants similarly observed that weaknesses in the organisation of the underlying knowledge would also constrain the ability of an AI system to retrieve it effectively.

These observations informed the later preparation of a dedicated SharePoint repository for the prototype. However, the two activities were analytically distinct. Document analysis examined the existing knowledge environment, whereas prototype data preparation involved selecting, cleaning, and organising a bounded set of approved sources for implementation.

The method therefore contributed both to problem understanding and to the assessment of knowledge readiness required by the artefact. The table distinguishes organisational knowledge sources from research-generated workshop material. Organisational sources were examined to understand the existing EA knowledge environment, while workshop material documented how this knowledge was mapped to stakeholder questions, ownership responsibilities, and prototype requirements.

Table 3.3. Document categories and their use in the study.

Document Category	**Examples**	**Purpose In the Study**
EA guidance	Guardrails, principles, architectural guidance	Examine available formal guidance and its suitability for retrieval
Start Assurance	Process description, self-assessment, agenda, minutes template	Understand the process and assess the selected pilot domain
Business capabilities	Capability information and ownership data	Examine capability-related knowledge needs and source availability
Governance material	Governance and engagement descriptions	Understand responsibility, decision boundaries, and escalation
Internal EA communication	SharePoint pages and presentations	Understand how EA knowledge is currently communicated

3.5.3 Workshop

One structured AI Use Cases and Knowledge Mapping Workshop was conducted on 19 May 2026. The meeting was scheduled for three and a half hours, from 13:00 to 16:30, and approximately three hours were devoted to the structured workshop activities. The workshop was designed, organised, and facilitated by the researcher. Members of the Enterprise Architecture team participated as subject-matter contributors, providing practical input concerning recurring EA questions, available knowledge sources, ownership responsibilities, risk, and the suitability of the identified use cases for further development.

The workshop was positioned between the interview-analysis stage and the artefact-design stage of the study. Its purpose was to validate and refine the preliminary use cases derived from the interviews and supplementary stakeholder discussions. The session was not intended as an open-ended brainstorming exercise. Instead, it was designed to transform broad use-case clusters into specific and actionable inputs for artefact design. These inputs included representative user questions, AI-task classifications, knowledge and document ownership, required knowledge sources, source-readiness observations, and candidate questions for the Minimum Viable Product.

Before the workshop, the researcher consolidated the interview-derived use cases into nine preliminary clusters. The researcher prepared a presentation explaining the workshop purpose and expected outputs, together with Excel worksheets for recording decisions and participant input. The prepared material distinguished broad use-case clusters from the more specific questions that users might direct to an EA guidance artefact. This distinction was important because the clusters were used only to organise the discussion, while the representative questions, ownership assignments, and supporting knowledge sources were treated as the actionable design inputs.

The researcher opened the session by presenting the findings derived from the stakeholder interviews and explaining how the use cases had been grouped into preliminary clusters. The researcher then introduced the workshop objectives, working principles, expected outputs, and six-phase structure. Throughout the session, the researcher guided the participants through the prepared worksheets, recorded their input, asked follow-up questions, clarified the distinction between broad opportunities and specific user questions, and kept the discussion focused on artefact requirements rather than general AI ideation or platform selection.

The workshop followed six phases:

1. **Use-case cluster validation:** Participants reviewed the preliminary clusters and considered whether each cluster represented a relevant EA support area. Each cluster could be retained and prioritised, merged with another cluster, reframed, or parked for later investigation.

2. **Question formulation:** Participants translated broad use-case descriptions into representative questions that stakeholders might direct to the EA function or an AI-supported guidance artefact. This phase moved the discussion from statements such as "AI should help with architecture" towards concrete questions such as "Do I need Start Assurance for this initiative?"
3. **AI-task classification:** The representative questions were classified according to the type of assistance expected from the artefact. The categories included finding information, explaining concepts, comparing alternatives, checking a proposal, summarising content, and routing the user to an appropriate human expert.
4. **Ownership assignment:** Participants considered three forms of responsibility: the knowledge owner, who understood the relevant subject area; the document owner, who knew where the supporting sources were located and whether they were current; and the validation owner, who could determine whether an AI-generated answer was acceptable.
5. **Knowledge-source mapping:** Each representative question was connected to the documents, SharePoint pages, models, templates, or other sources required to answer it. Participants considered whether the sources were current, authoritative, accessible, sufficiently complete, and suitable for AI-supported retrieval. Missing or outdated sources were recorded as knowledge-readiness gaps.
6. **MVP prioritisation:** Candidate question-and-document combinations were assessed according to value, frequency, input readiness, and risk. The intention was to identify use cases that offered practical value, occurred sufficiently often, had usable supporting information, and presented a manageable consequence if the generated answer was incomplete or incorrect.

Table 3.4. Workshop Phases and Outputs

Phase	Activity	Output
1	Validate use cases and clusters	Prioritised, merged, or parked use cases
2	Formulate representative questions	EA question bank
3	Classify AI-support tasks	Find, explain, compare, check, summarise, or route
4	Assign ownership	Knowledge, document, and validation owners
5	Map and assess sources	Required sources and readiness observations
6	Prioritise MVP candidates	Bounded pilot candidates

The researcher facilitated rather than predetermined the workshop decisions. The preliminary clusters and templates were prepared by the researcher, but the Enterprise Architecture team supplied the domain knowledge used to validate and refine them. Participants challenged the initial cluster structure, proposed alternative formulations, distinguished strategically important areas from lower-risk opportunities, identified required source material, and discussed situations in which human review remained necessary. The workshop therefore combined researcher-led structure with stakeholder-led validation.

The workshop generated several connected outputs. First, it produced a refined question bank representing realistic EA support needs. Second, it identified knowledge, document, and validation responsibilities for selected questions. Third, it connected the questions to available or required knowledge sources. Fourth, it revealed source-readiness problems, including outdated, fragmented, incomplete, or unavailable material. Finally, it produced a shortlist of candidate areas for the MVP and clarified which areas should be retained for later investigation.

After the workshop, the researcher consolidated the completed worksheets, meeting notes, and recorded discussion. The outputs were used to refine the use-case clusters, develop the artefact design requirements, assess knowledge readiness, and support the selection of Start Assurance as the principal pilot domain, with document discovery as a supporting capability. The workshop was therefore both a validation activity and a design-input mechanism within the Design Science Research process.

The workshop also had methodological limitations. Participation varied according to individual availability, and a separate attendance register was not maintained. The thesis therefore does not report an exact participant count. The workshop was facilitated by the researcher who had previously conducted the interviews and developed the preliminary clusters, which may have influenced the framing of the discussion. To reduce this risk, participants were explicitly invited to challenge, merge, reframe, deprioritise, or add to the proposed clusters. The workshop findings were also interpreted together with the interview evidence and document analysis rather than treated as an independent or statistically representative dataset.

3.6 Data Analysis

The data analysis process aimed to transform a large collection of qualitative data into actionable insights capable of informing the design of the proposed AI-supported Enterprise Architecture artefact. The analysis was conducted iteratively and formed a central component of the Design Science Research process, supporting both problem investigation and artefact development. The qualitative material was analysed using thematic analysis, which supports the systematic identification and interpretation of patterns across a dataset (Braun and Clarke 2006).

Data generated through interviews, document analysis, workshop discussions, workshop outputs, and researcher observations were analysed using a thematic analysis approach. Thematic analysis was considered appropriate because it enables the systematic identification, organisation, and interpretation of patterns across qualitative data while maintaining a strong connection to participants' experiences and perspectives.

The analysis process was conducted in three stages. First, interview transcripts and supporting material were subjected to open coding to identify meaningful observations and recurring concepts. Second, related codes were grouped into broader themes representing common organisational challenges and stakeholder needs. Finally, the identified themes were translated into design requirements that informed the architecture, functionality, and governance mechanisms of the proposed artefact.

This iterative process ensured traceability between stakeholder concerns, empirical findings, and the resulting design decisions. Rather than defining solution requirements based solely on theoretical assumptions, the analysis enabled requirements to emerge directly from organisational needs and validated stakeholder experiences.

3.6.1 Open Coding

The study employed thematic analysis, with open coding used during the initial coding stage. Open coding was used to generate descriptive labels close to the interview material, after which related codes were consolidated into higher-level themes and design-oriented use-case clusters.

The first stage of analysis involved open coding of the interview material. The purpose was to examine the qualitative evidence systematically and identify concepts, concerns, activities, actors, knowledge sources, and information needs expressed by participants. This approach enabled the researcher to remain closely connected to the empirical material while avoiding the premature application of the final themes.

The initial codes were kept close to the participants' wording. New extracts were compared with previously coded material so that similar observations could receive consistent codes. Codes were refined, merged, separated, or renamed when differences in meaning became apparent. For example, statements concerning difficulty locating documentation across SharePoint, Teams, presentations, and other repositories were initially coded separately and were subsequently grouped under a broader category relating to fragmented knowledge access.

Table 3.5 illustrates the open-coding procedure using anonymised extracts from the empirical material. The extracts were selected to demonstrate how individual statements were divided into meaningful fragments and translated into initial codes and provisional categories.

Table 3.5. Examples of the Open-Coding Procedure

Anonymised empirical extract	Initial open codes	Provisional category
Participant, architecture-related role: “It is difficult to find your way among all documents [1]… we don’t even ourselves find the way to our own documentation [2].”	[1] Difficulty navigating documents; [2] EA team unable to locate its own material	EA knowledge discovery and navigation
Participant, stakeholder role: “Knowledge is available on different platforms [1]… SharePoint sites and Teams, emails, and PPTs [2].”	[1] Knowledge distributed across platforms; [2] Multiple repository types	Fragmented knowledge sources
Participant, integration-related role: “Rules are clear [1]… but what is missing is how to practically use them [2].”	[1] Formal rules available; [2] Practical application unclear	Guardrail interpretation and application
Participant, architecture-related role: “What kind of guardrails to have in mind when we do that change [1]… to adhere to the EA principles [2]?”	[1] Identifying relevant guardrails during change; [2] Need to demonstrate alignment with EA principles	Contextual architectural guidance
Participant, business-architecture-related role: “This is information people very often request [1] because they need to know whom to talk to [2].”	[1] Recurring information request; [2] Need to identify the responsible owner	Business capability ownership and navigation
Participant, integration-related role: “It took almost a month [1] to understand we already had this integration [2].”	[1] Extended discovery effort; [2] Existing integration not visible	Integration discovery and reuse
Participant, product- or solution-related role: “What exactly comes from enterprise architecture [1]… that is unclear [2].”	[1] Unclear contribution of EA; [2] Uncertain responsibility boundary	EA governance and decision boundaries
Participant, product- or solution-related role: “What we should have mandate to decide on ourselves [1] and what should come from enterprise level [2]… it’s not quite clear [3].”	[1] Unclear local decision mandate; [2] Unclear enterprise-level responsibility; [3] Governance boundary ambiguity	EA governance and decision rights

Recurring initial codes included difficulty navigating EA documentation, knowledge distributed across repositories, dependence on informal expert support, limited practical explanation of guardrails, uncertainty concerning decision responsibilities, difficulty identifying capability owners, limited visibility of existing integrations, and the need for early risk-related guidance. Similar observations appeared in material associated with different stakeholder roles and organisational functions. The codebook nevertheless preserved distinctions where superficially similar statements concerned different underlying problems. For example, difficulty locating a guardrail was coded separately from difficulty applying the guardrail in a specific situation.

The coding process remained intentionally broad during the initial analysis. Similar observations were coded consistently regardless of the participant's role, while the source and role category were retained to preserve contextual differences. As additional material was examined, previously developed codes were compared with new extracts and refined, combined, separated, or renamed where necessary.

The initial codes were subsequently grouped into provisional categories reflecting recurring forms of EA knowledge-access difficulty and stakeholder support need. These categories included knowledge discovery, onboarding and learning, guardrail interpretation, Start Assurance support, business-capability navigation, capability-to-system relationships, integration discovery, governance boundaries, and security-related guidance. The categories provided the empirical basis for the higher-level themes and use-case clusters developed during the next stage of analysis.

Through open coding, the researcher established a structured and traceable connection between participant statements and the later analytical categories. The resulting categories were subsequently refined through comparison with the document-analysis findings and stakeholder-workshop evidence.

3.6.2 Theme Development

Following the completion of open coding, the identified codes were analysed to identify broader patterns and relationships. Related codes were grouped together based on conceptual similarity, resulting in the development of themes that reflected common challenges and opportunities associated with Enterprise Architecture knowledge accessibility.

The goal of this phase was not merely to categorise observations but to understand the underlying organisational issues represented by stakeholder experiences. Codes relating to document retrieval, repository navigation, and information discovery were therefore grouped into broader themes concerning knowledge discoverability. Similarly, codes associated with

learning, onboarding, architectural understanding, and organisational awareness were consolidated into themes related to knowledge internalisation and capability development.

The analysis resulted in the identification of nine major use case clusters representing distinct areas in which AI-supported assistance could potentially create value. These included Enterprise Architecture knowledge discovery, onboarding and learning support, guardrail interpretation, architecture review support, business capability navigation, capability-to-system mapping, integration discovery, governance support, and security architecture assistance. The clusters reflected recurring needs expressed by multiple stakeholders and served as an important bridge between problem analysis and solution design.

The preliminary themes were subsequently brought into stakeholder workshop for validation and refinement. During these session, participants reviewed the emerging clusters and evaluated their relevance, organisational value, and suitability for inclusion within an initial Minimum Viable Product. Stakeholders were encouraged to challenge, merge, prioritise, or deprioritise themes based on practical considerations and business relevance. Through this collaborative process, certain clusters were prioritised for immediate investigation, while others were considered strategically important but less suitable for an initial implementation.

The theme development process therefore combined researcher interpretation with stakeholder validation, increasing confidence that the final themes accurately represented organisational realities and practical requirements.

3.6.3 Requirement Extraction

The final stage of analysis focused on translating the validated themes into concrete design requirements for the proposed artefact. Within Design Science Research, this stage is essential because it establishes the connection between empirical findings and solution development.

Each validated theme was examined to identify the underlying capabilities that an AI-supported Enterprise Architecture assistant would need to provide. Rather than directly converting use cases into system features, the analysis sought to uncover the broader stakeholder requirements responsible for generating those use cases.

For example, use cases related to document retrieval, repository navigation, and knowledge discovery revealed a requirement for unified access to distributed knowledge sources. Similarly, onboarding and learning-related themes highlighted a requirement for explanation-oriented support capable of adapting responses to users with different levels of Enterprise Architecture knowledge. Themes associated with guardrail interpretation and governance support revealed a need for contextual reasoning capabilities capable of explaining not only architectural recommendations but also the rationale behind them.

The workshop played a particularly important role in this phase. Through question-generation activities, document-mapping exercises, ownership discussions, and MVP prioritisation sessions, stakeholders helped clarify the characteristics expected of a future AI-supported solution. These discussions highlighted the importance of source-grounded responses, knowledge ownership, validation mechanisms, transparent reasoning, escalation paths, and human oversight for higher-risk recommendations.

As a result, the final requirements extended beyond purely functional capabilities and included governance-related requirements intended to promote trust, accountability, and responsible use. The extracted requirements ultimately informed the design objectives, architecture selection process, and prototype development activities described in subsequent chapters.

The outcome of this analytical process was therefore not only a deeper understanding of Enterprise Architecture knowledge challenges but also a structured set of stakeholder-derived requirements that provided the foundation for artefact design and evaluation.

3.7 Artefact Design Process

Following the problem-identification and objective-definition stages of the Design Science Research Methodology, the next activity involved the design of an artefact intended to address the knowledge-accessibility challenges identified within the Enterprise Architecture environment of Sandvik Coromant. The purpose of the artefact-design process was not to implement a predetermined technology solution, but to develop a conceptually grounded and organisationally relevant approach capable of supporting stakeholders in locating, understanding, and applying Enterprise Architecture knowledge.

The design process was informed by three primary sources of evidence. First, the interview study identified recurring challenges related to knowledge discovery, onboarding, guardrail interpretation, process guidance, capability ownership, governance understanding, and dependence on architectural experts. Second, the document analysis provided insight into the structure, availability, and quality of the knowledge sources that would potentially support AI-assisted guidance. Third, the workshop activities validated and refined the initial findings by transforming broad opportunity areas into concrete user questions, mapping required information sources, assigning ownership responsibilities, and evaluating the feasibility of different use-case areas.

Rather than designing an artefact directly from individual user requests, the study adopted a progressive design approach in which empirical findings were synthesised into a set of design requirements. This ensured that the resulting artefact addressed recurring organisational needs rather than isolated stakeholder preferences.

3.7.1 From Findings to Design Requirements

The first step of the design process involved analysing the interview-derived clusters and workshop outcomes to identify recurring patterns that could inform artefact requirements.

The interview analysis revealed that stakeholders consistently experienced difficulties locating relevant Enterprise Architecture information across multiple repositories and documentation sources. Participants described challenges associated with discovering architectural artefacts, identifying the latest version of guidance documents, locating capability information, and understanding where specific forms of knowledge were maintained. Knowledge discovery therefore emerged as a fundamental requirement for any proposed solution.

A second recurring issue concerned interpretation rather than retrieval. Stakeholders frequently indicated that locating a document was often insufficient because additional contextual understanding was required to understand how architectural principles, guardrails, or governance decisions applied within a particular situation. These findings suggested that the artefact should support explanation and contextual guidance in addition to information retrieval.

The interview material also highlighted a strong dependence on experienced architects. Many participants relied on specific individuals to provide explanations, clarify ownership responsibilities, or identify appropriate information sources. This suggested a requirement for first-line guidance that could reduce routine questions while preserving escalation to human experts for situations requiring interpretation or judgement.

Additional requirements emerged from the workshop. Participants repeatedly emphasised the importance of authoritative knowledge sources, traceability, human oversight, and clear source ownership. Questions concerning guardrails and governance were considered particularly sensitive and were identified as requiring stronger validation and escalation mechanisms. This indicated that trust, transparency, and governance should be incorporated as core design considerations.

Collectively, these findings were consolidated into a set of design requirements that guided subsequent design activities. The resulting requirements served as evaluation criteria throughout the remainder of the design process.

The requirement for controlled retrieval across approved EA knowledge sources does not imply the duplication of information across repositories. Instead, it refers to the artefact's ability to retrieve complementary information from selected and authoritative sources while avoiding outdated or conflicting copies. For example, Start Assurance guidance may require information from the process description, self-assessment material, and relevant guardrails, as these sources provide different but related forms of knowledge.

Table 3.6. Derivation of Artefact Design Requirements from Empirical Findings

Identified Need	Design Requirement
Difficulty locating information	Unified knowledge retrieval
Fragmented repositories	Controlled retrieval across selected authoritative sources
Need for contextual interpretation	Conversational explanation capability
Dependence on experts	First-line guidance support
Unclear ownership responsibilities	Ownership visibility
Trust concerns	Source-grounded responses
Governance sensitivity	Human-in-the-loop review
Evolving documentation	Maintainable knowledge sources

The resulting design requirements informed the solution-evaluation criteria presented in Section 6.3, where they were combined with the architectural and technical considerations established in the original thesis assignment.

3.7.2 Identification and Assessment of Solution Alternatives

Once design requirements had been established, candidate AI approaches were examined to determine their suitability for supporting Enterprise Architecture knowledge retrieval and guidance.

The objective was not to identify the technically most advanced solution, but rather to identify an approach capable of satisfying organisational requirements relating to accessibility, governance, maintainability, trustworthiness, and practical adoption.

Several alternative approaches were considered during the study. These included enterprise copilot platforms, retrieval-augmented generation architectures, graph-based retrieval approaches, and open-source language model solutions. Each alternative was analysed against the requirements derived from the empirical material and evaluated according to factors such

as source grounding, organisational fit, governance support, maintenance effort, scalability, and implementation feasibility.

The assessment revealed that different approaches offered different strengths. Retrieval-focused architectures provided stronger control over organisational knowledge sources, while enterprise copilots offered advantages related to integration and adoption. More advanced graph-based approaches provided opportunities for modelling relationships between architectural artefacts but required greater preparation of underlying knowledge structures. Conversely, some open-source alternatives offered increased flexibility but introduced additional complexity related to governance, maintenance, and deployment.

Rather than selecting a solution solely on technical capability, the assessment focused on identifying an approach that could realistically operate within the organisational context while satisfying the requirements identified during the problem-investigation phase.

3.7.3 Pilot Domain Selection

Because Enterprise Architecture knowledge encompasses a broad range of topics and artefacts, it was necessary to define a bounded pilot scope for the artefact.

The workshop played an important role in this activity by providing a structured process for prioritising candidate use cases. Participants validated the interview-derived clusters, transformed them into concrete user questions, and evaluated them according to value, frequency, document readiness, and risk. The workshop outcomes demonstrated that not all use-case areas were equally suitable for an initial implementation.

The prioritisation exercise resulted in different outcomes for different clusters. Knowledge discovery, onboarding, guardrail interpretation, Start Assurance, and business-capability-related questions were retained as priority areas, while capability-to-system analysis, integration discovery, and security-related use cases were parked for future development.

Workshop participants also evaluated document availability and ownership readiness. Thirteen questions were classified as ready for use, four were identified as lacking sufficient supporting information, and one required document cleanup before being considered suitable for AI-supported retrieval. Key gaps included documentation relating to guardrail exceptions, governance concepts, capability-system relationships, and engagement-model guidance.

The workshop outcomes indicated that Start Assurance-related support represented a particularly attractive pilot candidate. Participants identified Start Assurance as a recurring organisational activity with clear user demand, identifiable documentation, manageable risk, and significant potential to reduce repetitive support requests. Document discovery was also

considered an important candidate because it addressed one of the most frequently reported problems across stakeholder groups.

Consequently, Start Assurance was selected as the principal pilot domain, with document discovery incorporated as a supporting capability.

3.7.4 Conceptual Artefact Design

The final stage of the design process involved translating stakeholder requirements and prioritised use cases into a conceptual artefact design.

At a conceptual level, the artefact was envisioned as an AI-supported Enterprise Architecture guidance assistant capable of interacting with users through natural language. Rather than functioning as a simple search interface, the artefact was designed to combine retrieval, explanation, guidance, and escalation capabilities.

The intended interaction model allows users to pose questions in natural language related to Enterprise Architecture processes, documents, capabilities, guardrails, and governance topics. Relevant information is retrieved from approved organisational knowledge sources and used to generate contextualised responses. Where responses involve uncertainty, incomplete information, or high-risk guidance, the artefact is designed to provide source references, communicate limitations, and direct users towards appropriate experts.

A key design principle emerging from the study was that the artefact should augment rather than replace Enterprise Architecture expertise. Consequently, the artefact was conceived as a first-line support mechanism intended to improve knowledge accessibility, reduce routine information-seeking activities, and support stakeholder understanding while maintaining human oversight for interpretation-intensive decisions.

The resulting conceptual design therefore reflects the intersection of the empirical findings, organisational requirements, and technological opportunities identified throughout the study. It also establishes a traceable connection between the problem identified in the case organisation and the artefact development activities presented in the following chapter.

3.8 Evaluation Approach

The evaluation approach in this study is formative and context sensitive. Rather than assessing artefact performance solely through technical measures, the evaluation focuses on whether the proposed artefact addresses the identified Enterprise Architecture knowledge-accessibility challenges in a meaningful way. The assessment therefore considers the relevance of the solution concept, its perceived usefulness for supporting Enterprise Architecture guidance, its

alignment with stakeholder needs, and its fit with organisational considerations such as governance, maintainability, and trustworthiness.

Consistent with the Design Science Research approach adopted in this study, evaluation is conducted in relation to the design requirements derived from interviews, document analysis, and workshop activities. The objective is to determine whether the proposed artefact demonstrates the potential to improve access to Enterprise Architecture knowledge, support stakeholder understanding, and reduce dependence on individual experts for routine guidance.

The evaluation is closely linked to the prototype and pilot-oriented nature of the thesis work. Stakeholder feedback plays an important role because the usefulness of an Enterprise Architecture guidance artefact depends not only on technical feasibility but also on whether intended users perceive it as accurate, understandable, and practically supportive. Evaluation in this study should therefore be understood as a combination of artefact-oriented assessment and qualitative validation within the context of the case organisation. The detailed evaluation activities and findings are presented in Chapter 7.

3.9 Ethical Considerations

Ethical considerations were relevant throughout the study due to its use of organisational documents, stakeholder perspectives, and an industrial case setting. Participation in interviews and workshop activities was voluntary, and participants were informed about the purpose of the study and how the collected material would be used. Care was taken to ensure that sensitive organisational information, confidential technical details, and personally identifiable information were not disclosed in the thesis.

Ethical considerations were also relevant to the AI-related aspects of the study. Since the proposed artefact is intended to support access to Enterprise Architecture knowledge, issues relating to data access, privacy, trust, and answer reliability were considered during the design process. The artefact is therefore positioned as a support mechanism that assists stakeholders in accessing and understanding relevant knowledge rather than replacing human architectural judgement. Consequently, governance, human oversight, and responsible maintenance of knowledge sources remain important considerations for the effective use of AI-supported guidance within the organisation.

3.10 Research Quality Considerations

The quality of the research was supported through the use of multiple sources of evidence, including interviews, document analysis, and workshop activities. The combination of these methods enabled triangulation and provided a richer understanding of the Enterprise

Architecture knowledge-accessibility challenges investigated in the study. The use of semi-structured interviews ensured consistent coverage of key topics while allowing participants to elaborate on issues relevant to their roles and experiences. In addition, thematic analysis provided a systematic approach for identifying patterns within the qualitative material and translating them into design requirements.

Research quality also depends on transparency in how the study progresses from problem identification to artefact design. For this reason, the thesis explicitly links empirical findings, identified themes, design requirements, and artefact decisions throughout the Design Science Research process. This traceability provides a clear rationale for how the proposed artefact was derived from the organisational context rather than from predetermined technological assumptions.

At the same time, the study recognises the limitations associated with a single-case design and the bounded scope of a master's thesis project. The intention is therefore not to produce universally generalisable conclusions, but rather to provide a credible and contextually grounded contribution that may inform both the case organisation and future research on AI-supported Enterprise Architecture guidance.

4. Case context and current Enterprise Architecture practice

4.1 Organisation Overview

This study was conducted at Sandvik Coromant, a business area of Sandvik, a global engineering group providing products, services, and digital solutions for the manufacturing, mining, and infrastructure industries. According to Sandvik's 2025 annual reporting, the group employed approximately 42,000 people, generated revenues of approximately SEK 121 billion, and operated in more than 150 countries. Within this broader context, Sandvik Coromant specialises in metal-cutting tools, machining solutions, manufacturing expertise, and digital capabilities supporting industrial operations.

The organisational setting is particularly relevant to this study because Sandvik Coromant operates in a knowledge-intensive and globally distributed environment. Information, standards, processes, and technology decisions need to be coordinated across organisational boundaries, functions, and geographical locations. Such an environment creates a growing need for shared knowledge, consistent guidance, and effective mechanisms for making organisational knowledge accessible to different stakeholder groups.

The company is also actively engaged in digital transformation initiatives, resulting in an expanding landscape of business, process, information, and technology-related knowledge. As architectural standards, capability models, governance structures, and decision-making practices evolve, employees increasingly require access to reliable and understandable guidance. Consequently, Enterprise Architecture plays an important role in supporting organisational alignment and helping stakeholders navigate complex information environments.

The present study was conducted in collaboration with the Enterprise Architecture team at Sandvik Coromant. The research emerged from an organisational interest in exploring how Artificial Intelligence could improve access to Enterprise Architecture knowledge and support stakeholders in locating, understanding, and applying architecture-related guidance. The case organisation therefore provides a suitable environment for investigating AI-supported knowledge retrieval and guidance within a real industrial setting.

4.2 Enterprise Architecture Function

The Enterprise Architecture function operates as a coordinating and enabling capability within Sandvik Coromant. Rather than focusing solely on technology planning, the function supports communication, organisational alignment, governance, and decision-making across business and technology domains.

The empirical investigation indicated that the Enterprise Architecture function is involved in maintaining architectural guidance, supporting governance processes, defining guardrails, developing capability models, facilitating architectural reviews, contributing to transformation initiatives, and helping stakeholders interpret organisational standards and expectations. Interview participants frequently described the EA team as a source of expertise consulted when clarification was required regarding architectural processes, responsibilities, or decisions.

The Enterprise Architecture function therefore performs both strategic and operational roles. At the strategic level, it contributes to architectural direction, governance, target architectures, and organisational alignment. At the operational level, it supports stakeholders seeking information about capabilities, standards, ownership responsibilities, architectural processes, and decision-making practices. This support role was particularly relevant to the study because many of the identified challenges concerned the accessibility and interpretation of existing architectural knowledge rather than the complete absence of such knowledge.

The formal EA team operates within a broader structure involving executive leadership, architectural governance bodies, and architecture roles distributed across the organisation. The Head of Enterprise Architecture and the EA team provide a central coordination point between the CIO, the Architecture Forum, the Architecture Community of Practice, and related governance bodies. Figure 4.1 provides a simplified representation of this governance and collaboration structure.

Figure 4.1 distinguishes the EA team from the wider architecture community. The central EA function is connected to the CIO and interacts with governance bodies responsible for enterprise processes and capabilities. The Architecture Forum provides a setting for architectural decision-making, recommendations, and the review of deliverables such as guidance material, blueprints, and guardrails. Internal organisational material describes the forum as involving architecture leaders representing different parts of the organisation and following an established governance and decision-making process.

The Architecture Community of Practice has a broader professional-development and collaboration role. It connects the EA team with functional lead solution architects, IT platform lead solution architects, major-initiative architects, and the wider Sandvik Coromant solution-architecture community. Current internal material distinguishes architecture work across three levels: enterprise-level direction, product-area consistency, and solution-level realisation. These levels represent different scopes and planning horizons rather than completely separate architecture professions. The structure includes the central EA function, architecture forums, communities of practice, distributed architect roles, and related governance bodies. The relationships represent coordination and governance interactions rather than uniform line-management relationships.

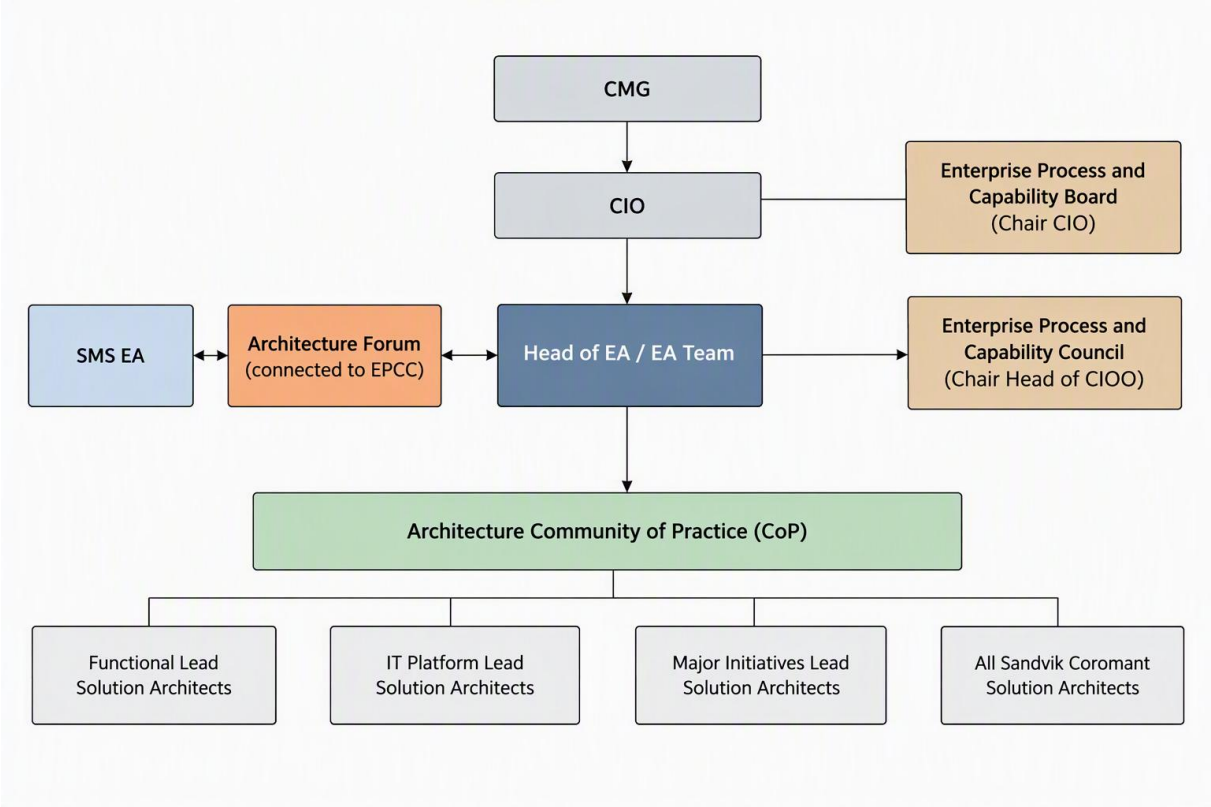


Figure 4.1 Enterprise Architecture Governance and Collaboration Structure

The relationships shown in Figure 4.1 should not be interpreted entirely as formal reporting lines. The figure primarily represents governance, coordination, and professional collaboration. The broader architecture community extends the reach of the central EA team by connecting enterprise-wide direction with area-level architecture and solution-level implementation.

4.3 Enterprise Architecture Knowledge Artefacts

The EA knowledge environment included several artefacts with different purposes, ownership arrangements, and user groups. Some artefacts, such as the EA Charter, playbook, principles, and guardrails, provided common direction across the organisation. Others, including Start Assurance guidance, capability maps, roadmaps, conceptual architectures, and Architecture Decision Records, supported more specific planning, assurance, and decision-making activities. Figure 4.2 provides an analytical summary of the participant groups involved in developing, reviewing, maintaining, or using these artefacts.

EA knowledge artifacts \ Participants		Business EA	Information EA	Process EA	Technology EA	Architecture Forum	Area Architects	Solution Architects	Business Owners
Business	**Capability and value-stream models**	D	d	d/U	d/U	d	U	U	D/U
Process	Process frameworks and maps	d/U	d	D	u	u	U	U	D/U
Information	Information models and data principles	d	D	d/U	d/U	u	U	U	D/U
Technology	Application and platform landscape	d	d/U	u	u	d/U	D/U	U	u
Cross-domain	Principles and guardrails	d/U	d/U	d/U	D	D/U	U	U	n
	Start Architecture and Assurance	D	d/U	d/U	d/U	D/U	U	U	U
	Target and conceptual architectures	d/U	d/U	d/U	D	D/U	D/U	d/U	n
Planning	**Capability and transformation roadmaps**	D/U	d/U	d/U	d/U	d	d/U	d/U	D/U
Governance	Architecture decision records	d/U	d/U	d/U	d/U	D/U	d/U	d/U	n

Figure 4.2. Developers and Users of Core Enterprise Architecture Knowledge Artefacts

[D] Principal developer/accountable knowledge owner [d] Contributor or reviewer
[U] Regular user [u] Occasional or situational user Combined symbols indicate both roles

Figure 4.2 shows that some artefacts were associated primarily with one EA area. Business capability and value-stream models were most closely connected to Business EA, process frameworks and process maps to Process EA, information models and data principles to Information EA, and application and platform landscapes to Technology EA. However, these artefacts were not independent. Business capabilities were connected to processes, information, applications, and ownership, while process and information models supported the interpretation of how capabilities were realised.

Principles, guardrails, Start Architecture, Start Assurance, target architectures, roadmaps, and decision records had a more cross-domain character. These artefacts required input from different EA areas, architecture forums, product-area architects, solution architects, and relevant business owners. The workshop findings reinforced this distributed structure by mapping representative questions to knowledge sources and responsible experts rather than assuming that all EA information was developed and maintained by one central role.

The matrix also illustrates why EA knowledge access was not solely a document-retrieval problem. The usefulness of an artefact depended on knowing which source was authoritative, which architectural area could interpret it, and who could validate its application in a specific context. This distributed ownership informed the later requirements for source grounding, visible ownership, human review, and escalation to the appropriate expert.

4.4 Current Enterprise Architecture Support Context

EA-related support is provided through a combination of formal documentation, SharePoint-based knowledge, direct interaction with architects, and established governance and assurance activities. Stakeholders can consult available documents or engage members of the architecture function when additional explanation or contextual interpretation is required.

Start Assurance represents one example of this support context. It is an early architecture-guidance process intended to help initiatives and continuous improvements obtain an appropriate starting point while balancing short-term needs with Sandvik Coromant's strategic direction and Enterprise IT Guardrails. The process provides a high-level review of an initiative from a strategic and architectural perspective, resulting in documented findings, recommendations, and actions for its realisation.

The current Start Assurance routine begins with a self-assessment. The self-assessment is completed for new major initiatives or projects before a certain Decision Point and for relevant continuous-development initiatives or changes that may be affected by the Guiding Principles or Enterprise IT Guardrails. The responsible participants are typically the Solution Architect together with the Project Manager, Initiative Lead, or Application Product Owner.

The outcome of the self-assessment determines the following course of action. If all assessed areas are green, the completed self-assessment is sent to the Business Architecture team, and the process may conclude without a separate Start Assurance meeting. If one or more areas are non-green, the initiative team contacts the Business Architecture team or the relevant specialist function, such as Integration or Security. The identified issues are then discussed, relevant actions are considered, and a decision is made on whether an updated self-assessment is sufficient or whether a formal Start Assurance meeting is required.

When a meeting is required, the Business Architecture team schedules the session and invites representatives from the applicable functions. The initiative team prepares a pre-read using the Start Assurance agenda template and distributes it to the participants one week before the meeting. The pre-read provides the material required for the architectural discussion and is prepared by the Solution Architect together with the Project Manager, Initiative Lead, or Application Product Owner.

During the meeting, the initiative is reviewed using the prepared material and the completed self-assessment. The standard minutes template structures the discussion around the initiative presentation, business targets and scope, initiative objectives, dependencies and assumptions, the self-assessment, and the resulting conclusions and actions. The Business Architecture team facilitates the meeting and records the discussion using the Start Assurance minutes template.

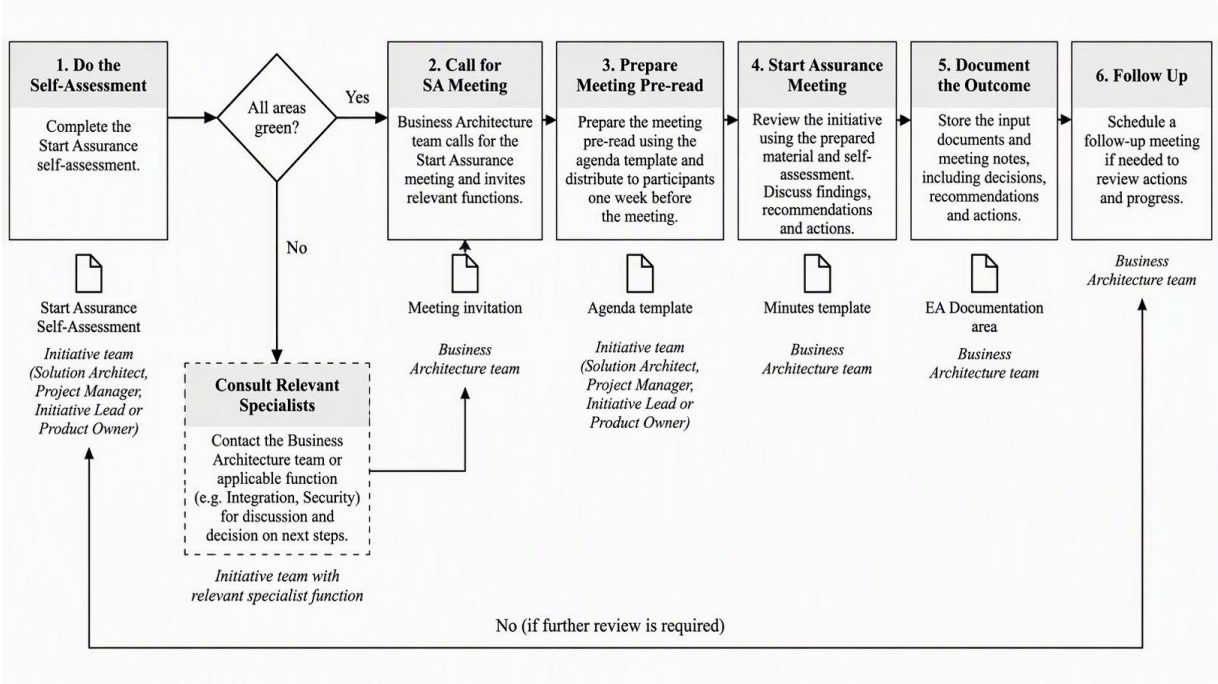


Figure 4.3. Current Start Assurance Process

After the meeting, the Business Architecture team stores the input documents and meeting notes, including the identified decisions, recommendations, actions, responsible parties, and relevant deadlines, in the EA documentation area. When unresolved questions or actions remain, a follow-up meeting may be scheduled. The process therefore extends beyond the meeting itself and creates a documented basis for subsequent architectural and implementation activities.

The proposed AI-supported artefact was not intended to replace this process or automate architectural decisions. Its intended role was to improve access to approved guidance, help users locate and understand the required material, support completion of the initial assessment, and provide structured first-line preparation before human involvement. Questions requiring contextual interpretation, specialist input, exception handling, or formal architectural judgement would continue to be reviewed by the responsible experts and governance bodies.

5. Empirical findings and problem analysis

5.1 Introduction

This chapter presents the findings from the interviews, document analysis, and workshop activities conducted during the problem-identification phase of the study. The purpose of the chapter is not only to describe the current situation within the Enterprise Architecture (EA) environment but also to analyse the factors affecting access to and use of EA knowledge. The findings subsequently provide the foundation for the design requirements that guide artefact development.

The analysis revealed that the primary challenge was not the absence of Enterprise Architecture knowledge. Instead, stakeholders experienced difficulties discovering, understanding, and applying existing knowledge distributed across documents, repositories, models, processes, and expert networks. The chapter therefore examines the current knowledge landscape, identifies recurring challenges, explores stakeholder needs, and concludes with the design requirements derived from the empirical investigation.

5.2 Enterprise Architecture Knowledge Landscape

The empirical investigation showed that Enterprise Architecture knowledge exists across a wide range of artefacts, repositories, and organisational practices. Participants described a knowledge environment consisting of architecture guardrails, principles, business capability models, governance artefacts, process descriptions, review processes, presentations, and supporting documentation. These artefacts are distributed across multiple locations, including SharePoint sites, Teams environments, presentations, and specialist repositories.

While this information was considered valuable, stakeholders frequently indicated that knowledge was difficult to navigate because relevant information was dispersed across multiple platforms. Several participants noted that locating the correct document often required prior knowledge of where the information was stored or who maintained it. Similar observations were also made during the workshop, where participants repeatedly linked knowledge accessibility challenges to the distributed nature of Enterprise Architecture information.

The findings therefore suggest that the challenge is not solely related to information availability but also to information accessibility. In many situations, knowledge exists within the organisation but remains difficult to access without assistance from architects or other knowledgeable individuals.

5.3 Challenges in Accessing and Using Enterprise Architecture Knowledge

5.3.1 Difficulty Discovering Relevant Information

One of the most frequently discussed challenges concerned the discovery of relevant information. Participants described situations in which they needed to search across SharePoint sites, Teams channels, PowerPoint presentations, and other repositories to locate architecture-related guidance. The interview material suggested that users often struggled to identify where information was stored and which version should be considered authoritative.

One interview participant described the knowledge environment as follows: “Knowledge is available on different platforms … SharePoint sites and Teams, emails, and PPTs.” Another participant explained that “it is difficult to find your way among all documents.”

These difficulties were also reflected in the workshop discussions, where document discovery emerged as one of the highest-priority opportunity areas for AI-supported assistance. Questions relating to the location of documents, capability maps, guardrails, and process descriptions were repeatedly identified as common and high-value support requests.

5.3.2 Dependence on Enterprise Architecture Experts

The investigation further revealed a strong dependence on individual experts as mediators of Enterprise Architecture knowledge. Stakeholders frequently relied on architects to explain concepts, identify relevant documents, clarify ownership responsibilities, and interpret architectural guidance. While this support was valuable, it also created bottlenecks and introduced scalability challenges as the volume of questions increased.

The recurring need to identify responsible experts was reflected in one participant’s observation that this was “information people very often request because they need to know whom to talk to.”

The onboarding-related findings were particularly illustrative of this challenge. Participants described situations in which understanding capability models, architectural concepts, and organisational structures required lengthy walkthroughs or direct interaction with experienced architects. The reliance on experts therefore extended beyond governance decisions and into day-to-day knowledge acquisition.

5.3.3 Challenges in Interpreting Architectural Guidance

The findings also suggested that locating information alone was often insufficient. Participants frequently emphasised the need to understand how architectural guidance should be applied in specific situations. This was particularly visible in discussions relating to Enterprise Architecture guardrails and architectural principles. Several participants explained that the existing guidance described what should be achieved but provided limited support regarding practical application and contextual interpretation.

One participant summarised the distinction between availability and practical usability: “Rules are clear … but what is missing is how to practically use them.”

Workshop discussions reinforced this observation. Participants highlighted that the usefulness of guidance depended on understanding context, exceptions, and practical examples rather than merely accessing policy documents. As a result, interpretation emerged as a distinct challenge separate from information retrieval.

5.3.4 Knowledge Quality and Governance Concerns

A further challenge concerned the quality and maintenance of knowledge sources. Participants noted that outdated content, duplicated artefacts, missing documentation, and unclear ownership could create confusion and reduce trust in retrieved information. These concerns became particularly visible during the document-readiness assessment conducted in the workshop.

The workshop identified several information gaps, including missing guidance relating to guardrail exceptions, governance artefacts, engagement-model concepts, and capability-system relationships. It also became apparent that successful AI-supported guidance depended not only on retrieval capabilities but also on the availability of trustworthy and well-maintained knowledge sources.

5.4 Stakeholder Needs for AI-Supported Enterprise Architecture Guidance

While the challenges described above highlighted limitations in the current environment, the empirical investigation also identified several recurring support needs that stakeholders believed could benefit from AI assistance.

The first category concerned knowledge discovery. Stakeholders frequently wanted support locating relevant documents, capability maps, standards, architectural guidance, and process descriptions. The need focused on reducing the effort required to navigate fragmented repositories and identify relevant sources.

A second category concerned onboarding and learning. Participants described the need to help new architects, new employees, and non-EA stakeholders understand Enterprise Architecture concepts, structures, and practices. AI-supported guidance was viewed as a potential mechanism for providing introductory explanations and supporting self-directed learning.

A third category involved interpretation support. Stakeholders expressed interest in guidance relating to Enterprise Architecture guardrails, capability relationships, governance structures, and architectural processes. Rather than simply locating documents, users wanted explanations that connected information to a specific situation or decision context.

Finally, process-oriented support emerged as a particularly important area. Questions relating to Start Assurance, architecture reviews, capability ownership, and governance responsibilities were repeatedly identified as recurring and valuable opportunities for AI-supported assistance.

5.5 Prioritisation of Enterprise Architecture Support Opportunities

The workshop provided a structured mechanism for validating and prioritising the interview-derived clusters. Participants assessed potential areas according to value, frequency, document readiness, and risk. The outcome demonstrated that not all opportunity areas were equally suitable for an initial implementation.

The validation process resulted in three outcomes: prioritisation, merging, and parking. A total of 18 use cases were retained as prioritised candidates, seven were merged into broader categories, and 16 were parked for future investigation. Clusters relating to knowledge discovery, onboarding, guardrail interpretation, Start Assurance, and business capabilities received the strongest support. Conversely, capability-to-system mapping, integration discovery, and security-related topics were recognised as relevant but considered less suitable for an initial pilot due to complexity, readiness concerns, or scope considerations.

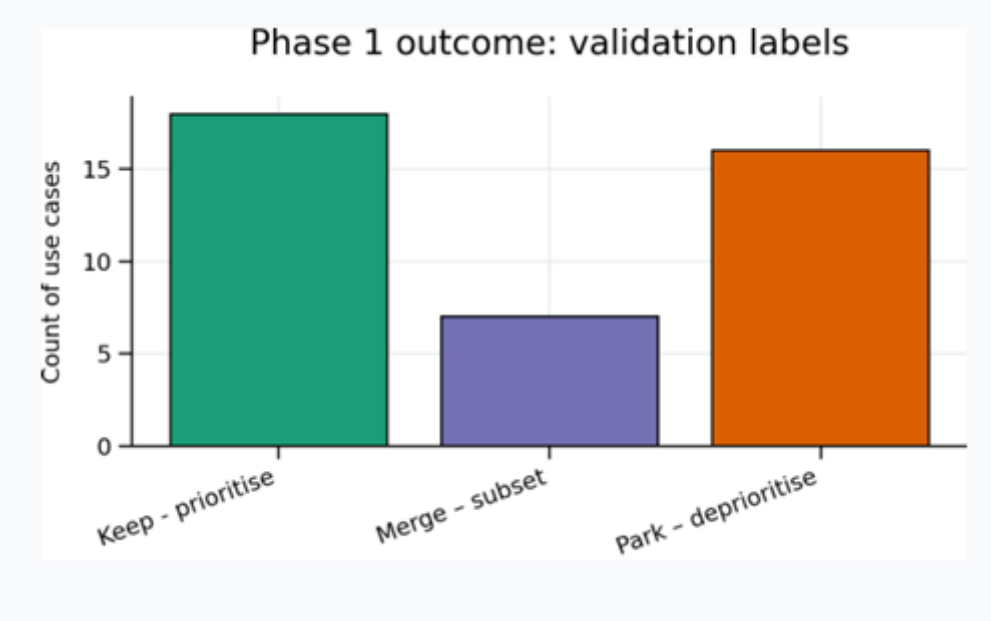


Figure 5.1. Workshop Validation Outcomes

The workshop therefore narrowed the design scope and helped identify areas capable of providing practical value while remaining feasible within the context of the study.

5.6 Knowledge Readiness Analysis

An important finding was that AI readiness depended strongly on knowledge readiness. To investigate this relationship, workshop participants mapped questions to required knowledge sources and assessed the quality and availability of supporting information.

The readiness assessment classified 13 questions as ready, four as missing required information, and one as requiring cleanup before use. Missing areas included guidance relating to guardrail exceptions, capability-to-system relationships, governance artefacts, and engagement-model information.

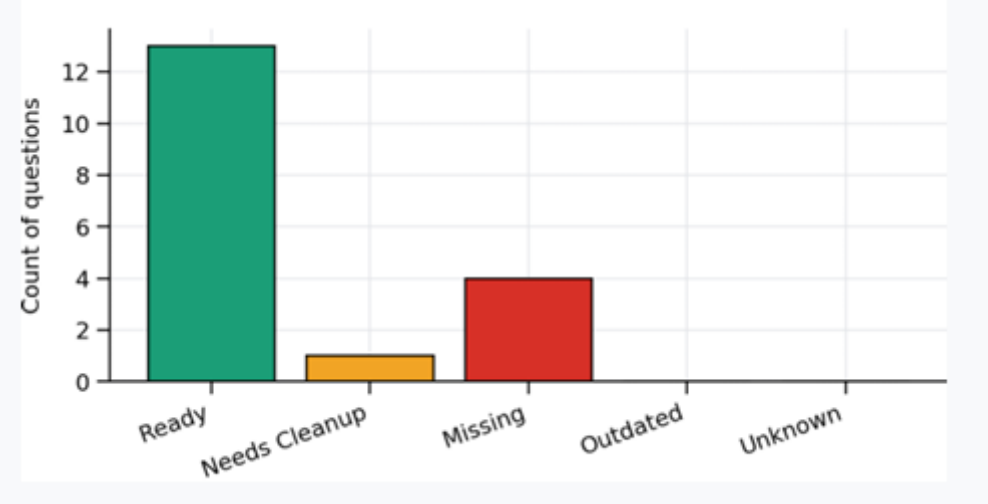


Figure 5.2. Document-Readiness Assessment

The findings demonstrated that the success of any AI-supported guidance solution depends not only on retrieval technologies but also on the quality, ownership, and maintenance of the underlying knowledge assets. This observation became an important consideration in the subsequent artefact-design process.

5.7 Design Requirements

The empirical findings were translated into a set of design requirements that informed the development of the proposed artefact.

First, the artefact should improve knowledge accessibility through natural-language interaction and support retrieval across distributed Enterprise Architecture knowledge sources. Second, it should provide contextual explanations rather than functioning solely as a document-search mechanism. Third, responses should be grounded in authoritative sources and provide visibility into where information originates. Fourth, the artefact should support escalation to human experts when knowledge is missing, or situations require judgement beyond available guidance. Finally, governance, ownership, and maintainability must be considered to ensure that organisational knowledge remains trustworthy and sustainable over time.

In addition, several cross-cutting requirements were consistently identified across interviews and workshop activities. These included context-aware clarification before answering, source-grounded responses, confidence awareness, human review of sensitive recommendations, and escalation mechanisms when reliable answers could not be provided.

Taken together, these requirements establish a direct link between the identified organisational challenges and the artefact developed in the following chapter. Rather than emerging from technological possibilities alone, the requirements are grounded in stakeholder experiences,

knowledge-accessibility needs, and organisational constraints identified throughout the empirical investigation.

5.8 Selection of Pilot Domain

The empirical investigation identified several areas in which AI-supported guidance could provide value within Enterprise Architecture practice. However, the study required a bounded pilot domain that could be designed and evaluated within the available scope. The pilot-selection process therefore focused on identifying a recurring EA support need that was organisationally relevant, supported by usable knowledge sources, and suitable for controlled AI assistance.

The workshop was designed to move from broad use-case clusters towards concrete and testable questions. The clusters were used to organise related areas of potential support, but they were not treated as the final unit of prioritisation. Instead, the workshop focused on the questions stakeholders might ask, the documents required to answer them, the ownership of the underlying knowledge, and the risks associated with an incorrect response.

The selection process followed a structured sequence. Preliminary use cases were first reviewed and classified as priorities, subsets of other use cases, candidates for later consideration, or possible new areas. The retained use cases were then translated into representative questions and classified according to the intended AI task, including finding information, explaining concepts, comparing alternatives, checking proposals, summarising content, or routing users to an appropriate expert. Knowledge, document, and validation ownership were subsequently considered before the required sources were mapped and assessed for readiness.

MVP candidates were selected from combinations of representative questions and supporting documents rather than from the use-case clusters alone. Four criteria were used:

- **Value:** whether the use case could improve efficiency or decision-making.
- **Frequency:** how often the corresponding question or support need occurred.
- **Input readiness:** whether the required documents were available and usable.
- **Risk:** the possible consequence of an incorrect or insufficiently contextualised response.

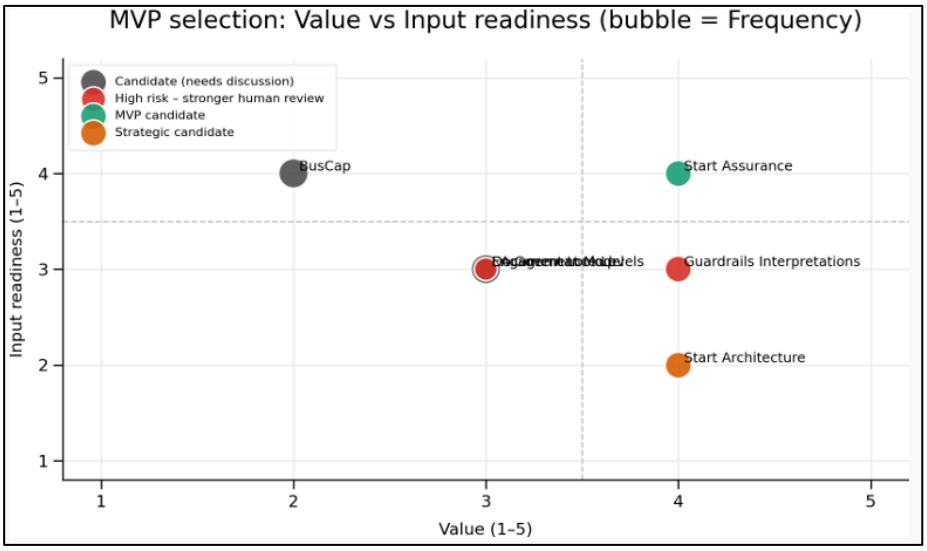


Figure 5.3. MVP Prioritisation of Candidate Use Cases

Each criterion was assessed on a five-point scale, where 1 represented a low level and 5 represented a high level. High value, frequency, and input-readiness scores were favourable, whereas a high risk score indicated a need for stronger validation, expert involvement, or restricted automation. The criteria were interpreted individually and were not aggregated into a single average.

As shown in Table 5.1, the assessment did not produce an unambiguous numerical winner. Document lookup and business capability lookup demonstrated strong potential because of their expected frequency, knowledge readiness, and comparatively low risk. However, these use cases primarily concerned locating information or ownership data and therefore provided a narrower basis for evaluating the proposed guidance artefact.

Guardrail interpretation was considered valuable and frequently required, but it received the highest risk score. The interpretation of a guardrail may depend on the context of an initiative, applicable exceptions, and professional architectural judgement. Guardrail interpretation was therefore retained as a relevant support area but was not considered suitable for autonomous guidance without validation or escalation.

Start Assurance received consistently high scores for value, frequency, and input readiness while retaining a comparatively manageable risk score. Within Sandvik Coromant, Start Assurance is an early architecture-guidance process intended to help initiatives and continuous improvements achieve an appropriate starting point. The process considers an initiative in relation to strategic direction and Enterprise IT Guardrails and results in documented advice, recommendations, and actions relevant to its realisation.

Table 5.1. Workshop Assessment of Candidate Pilot Use Cases

Candidate Use Case	Value	Frequency	Readiness	Risk
Start Assurance	4	4	4	2
Guardrail Interpretation	4	4	3	5
Document Lookup	3	5	3	1
Business Capability Lookup	2	5	4	1

Note: *Scores range from 1, low, to 5, high. High value, frequency, and input-readiness scores favour pilot selection, whereas a high-risk score indicates a greater need for validation and human oversight. The scores were used to structure comparison and were not combined into an overall average.*

The selection of Start Assurance was also consistent with the workshop's question-oriented approach. The workshop material used the representative question, "Do I need Start Assurance for this initiative?", as an example of sufficiently concrete problem formulation. This demonstrated how a broad use-case area could be converted into a question that required identifiable knowledge, a defined answer, and an assessment of the need for human involvement.

Start Assurance offered a broader basis for artefact development than the lookup use cases. It required the artefact to retrieve relevant knowledge, ask structured questions, provide process guidance, identify applicable documents, and recognise when engagement with the EA function remained necessary. It therefore allowed the study to examine both knowledge retrieval and guided interaction within one bounded organisational process.

Based on the workshop assessment and the objectives of the study, Start Assurance was selected as the principal pilot domain. The decision was not based on the numerical scores alone. It also reflected the workshop decision logic, according to which an MVP candidate should combine high value, usable input sources, and manageable risk. Questions associated with high consequences or insufficiently mature sources were instead treated as requiring further validation or later development.

Document discovery was incorporated as a supporting capability within the Start Assurance pilot rather than treated as a separate competing domain. Access to relevant process descriptions, self-assessment material, templates, guardrails, and supporting guidance was necessary throughout the Start Assurance interaction. The pilot consequently tested document retrieval within the context of a defined EA support process.

The remaining use cases were not rejected permanently. Guardrail interpretation and business capability lookup remained relevant candidates for future development. More complex areas, such as capability-to-system mapping, integration discovery, and security-related guidance, required greater source readiness, stronger validation arrangements, or more extensive specialist involvement before they could be considered suitable for broader AI-supported use.

The selected pilot therefore provided a bounded and organisationally relevant context for designing and evaluating the artefact. It connected recurring EA questions with identifiable knowledge sources, structured process guidance, and human escalation while preserving the responsibility of the EA function for formal architectural judgement.

6. Artefact Design and Development

6.1 Design Objectives

The findings presented in Chapter 5 demonstrated that the primary challenge within the Enterprise Architecture environment was not the absence of architectural knowledge, but the difficulty of accessing, understanding, and applying that knowledge in practice. Stakeholders reported recurring difficulties locating relevant information, interpreting architectural guidance, identifying ownership responsibilities, and navigating distributed knowledge sources. These challenges resulted in continued dependence on a limited number of experts and reduced the scalability of existing support mechanisms.

Consequently, the objective of the artefact was to improve access to Enterprise Architecture knowledge while preserving the governance structures and expert oversight required within an organisational setting. Rather than replacing Enterprise Architects, the proposed artefact was intended to function as a first-line guidance mechanism capable of assisting stakeholders with knowledge discovery, contextual explanation, and process-related support.

Based on the design requirements derived from the empirical investigation, the artefact was expected to:

- Improve discoverability of Enterprise Architecture knowledge.
- Provide natural language access to distributed knowledge sources.
- Support interpretation of architectural guidance through contextual explanations.
- Reduce routine dependency on Enterprise Architecture experts.
- Provide source-grounded responses and improve transparency.
- Support escalation when human expertise is required.
- Operate within organisational governance and security constraints.

These objectives subsequently guided the identification and evaluation of potential solution approaches.

6.2 Candidate AI Solution Approaches

The technical investigation considered several approaches for implementing AI-supported Enterprise Architecture guidance. The alternatives were not treated as interchangeable products. They represented different levels of abstraction, technical control, implementation effort, and organisational responsibility.

The initial research plan proposed a comparison of low-code enterprise assistants, custom RAG pipelines, open-source language models, and agentic workflows. It also stated that the preferred approach should be selected through a trade-off analysis rather than assumed at the beginning of the study.

Three principal approaches were retained for detailed consideration:

1. Microsoft Copilot Studio

2. A custom RAG architecture, including vector and graph-based retrieval

3. An open-source LLM-based architecture

The assessment focused on the suitability of each approach for the organisational problem established in Chapter 5. The relevant question was therefore not which approach possessed the greatest technical capability in absolute terms, but which approach provided an appropriate balance between response quality, organisational fit, governance, security, cost, and maintainability.

6.2.1 Microsoft Copilot Studio

Microsoft Copilot Studio is a low-code platform for creating and customising conversational agents. Internal platform guidance describes it as an end-to-end conversational AI environment that supports natural-language and graphical agent development, testing, publication, and integration with services such as Power Automate. Internal guidance further positions Copilot Studio as appropriate when a business requires a purpose-specific agent that answers questions from approved knowledge sources, supports a defined process, or provides a guided conversational experience.

The approach was relevant to the case organisation because much of the initial pilot knowledge was represented in Microsoft 365 and SharePoint-based content. Internal guidance identified Copilot Studio as a governed and shareable low-code option that can publish agents to Teams, Microsoft 365 Copilot, or the organisational agent store. Microsoft documentation also explains that Copilot Studio can ground published agents in configured knowledge sources, including SharePoint, Dataverse, uploaded documents, and enterprise data connected through supported

connectors. Access to authenticated sources can operate in the context of the individual user, limiting retrieved content to material that the user is authorised to access.

These characteristics aligned with several design requirements. Copilot Studio supported a familiar conversational interface, could make use of existing organisational content, and provided a comparatively direct route to building a bounded pilot. The platform was also suitable for combining retrieval with structured conversational topics and workflows. During the internal demonstration, the prototype showed how a Start Assurance interaction could collect user responses, validate inputs, branch according to the answers provided, and guide the user through a defined process. The same discussion identified possible workflow actions such as storing responses, sending information, and initiating follow-up activities.

The platform also offered advantages from a governance perspective. Internal guidance stated that Copilot Studio agents should use approved knowledge sources, operate in approved environments, and not be presented as authoritative truth. Microsoft documentation identifies controls for data policies, agent capabilities, knowledge sources, connectors, publication channels, monitoring, identity, and credit usage. These controls did not remove the need for artefact-specific governance, but they provided an organisationally recognised foundation on which such governance could be established.

Practical testing also revealed limitations. The quality of responses remained dependent on the selection and preparation of source content. Tests performed with the complete EA document environment were less satisfactory than tests conducted with a smaller and cleaned source set. The internal technical discussion attributed part of this limitation to restricted control over document chunking and relationships between information distributed across different files.

Copilot Studio therefore offered a strong fit for rapid development, Microsoft 365 integration, guided conversational flows, governance alignment, and organisational adoption. Its limitations were most visible where the use case required detailed control over retrieval, explicit modelling of relationships, systematic experimentation with retrieval strategies, or deeper customisation of the AI pipeline.

6.2.2 Retrieval-Augmented Generation Architecture

A custom Retrieval-Augmented Generation architecture separates information retrieval from response generation. Organisational content is prepared, divided into appropriate units, enriched where necessary, indexed, and retrieved in response to a user question. The selected passages are subsequently passed to a language model as contextual input for generating an answer.

The main advantage of this approach is the level of control it provides over the retrieval process. Microsoft's RAG design guidance describes a pipeline in which source material is processed

through chunking, metadata enrichment, embedding, indexing, retrieval, packaging of relevant results, and response generation. The guidance also emphasises that the design of chunking, enrichment, search, indexing, and evaluation requires explicit choices rather than being a single automatic configuration.

This additional control was relevant to the EA context because architectural information frequently depends on relationships across artefacts. A guardrail may refer to a broader principle, a capability may connect to systems and ownership information, and process guidance may depend on several documents that were not written as a single knowledge unit. A custom retrieval architecture makes it possible to experiment with how such information is divided, represented, connected, ranked, and presented to the model.

The technical investigation considered both vector-based and graph-oriented retrieval. Vector retrieval represents document passages through embeddings and identifies content on the basis of semantic similarity. This approach is suitable where the objective is to find passages that are conceptually related to a natural-language question, even when the wording differs.

Graph-oriented retrieval offers a different form of support by representing entities and relationships explicitly. During the later technical phase, entities were extracted from selected SharePoint-based PDF content and used to explore a knowledge graph in Neo4j Community. The proposed evaluation plan compared a Copilot-based agent with vector and graph retrieval agents in Microsoft Foundry, using criteria such as setup complexity, grounding, hallucination, and cost.

A custom RAG architecture also supports more deliberate orchestration. The design can classify a query before invoking retrieval or generation, determine whether the request is factual, contextual, or outside the defined scope, and avoid unnecessary model calls. An internal demonstration showed this form of workflow as a possible method of reducing token consumption while improving control over when the model is used.

At the same time, a custom RAG approach transfers additional responsibility to the implementing organisation. The solution requires decisions regarding data ingestion, chunking, embeddings, indexing, retrieval configuration, security, model selection, orchestration, monitoring, evaluation, infrastructure, and ongoing operational ownership. Microsoft's documentation notes that enterprise RAG must address complex queries, access to multiple sources, token constraints, response time, and granular security controls.

The custom RAG approach therefore provided the strongest basis for retrieval experimentation and long-term architectural control. However, it also required greater engineering capacity and a more mature operating model than the low-code alternative. Internal guidance similarly distinguished Microsoft Foundry from Copilot Studio by describing Foundry as appropriate

when model choice, custom pipelines, reusable endpoints, multi-agent workflows, deployment control, and monitoring justify the additional engineering responsibility.

6.2.3 Open-Source LLM Approach

A third alternative involved the use of open-source Large Language Models deployed within a custom architecture.

Open-source approaches offer substantial flexibility regarding model selection, customisation, deployment strategy, and experimental design. They also provide opportunities to investigate advanced AI capabilities without being constrained by vendor-specific platforms.

From a research perspective, open-source solutions provide valuable opportunities for experimentation and technical innovation. However, they also introduce additional requirements relating to infrastructure, security management, operational support, governance, and maintenance.

For the present study, the open-source alternative was therefore treated as a technically relevant comparison rather than the preferred path for the bounded MVP. The thesis scope did not include establishing the full operational and governance capability required for a self-managed enterprise AI platform. The approach remained valuable as a future option where data sovereignty, model control, or specialised performance requirements justify the additional organisational responsibility.

6.3 Decision Criteria for Solution Selection

The candidate solution approaches were evaluated using criteria derived from three sources. First, the original thesis assignment established the need to examine architectural integrity, data sovereignty, maintainability, and cost in relation to performance. Second, the empirical investigation introduced organisational requirements concerning source grounding, trust, governance, human oversight, and knowledge ownership. Third, the technical investigation identified retrieval control and relationship-aware reasoning as relevant to EA questions requiring information from several related artefacts.

The criteria were therefore neither predetermined entirely at the beginning of the study nor derived solely from the workshop. Instead, they were developed progressively by combining the initial project requirements with the stakeholder-derived design requirements presented in Section 3.7.1 and the observations made during technical experimentation.

Table 6.1. Criteria Used to Evaluate Candidate Solution Approaches

Criterion	Reason for inclusion
Accuracy and relevance	The solution should retrieve information that addresses the user's question and avoid presenting unrelated content as EA guidance.
Grounding and source transparency	Generated responses should be connected to approved EA knowledge sources and allow users to identify the basis of the answer.
Relationship-aware reasoning	Some EA questions require connections to be identified across capabilities, guardrails, processes, systems, owners, and other architectural artefacts.
Governance	The solution should support clearly defined ownership, validation, controlled publication, and accountable use.
Security and access control	The solution should preserve organisational access restrictions and protect internal EA knowledge.
Human oversight	Ambiguous, sensitive, or judgement-intensive questions should be escalated for expert review.
Maintainability	Knowledge sources, workflows, system instructions, and technical components should remain manageable after the thesis.
Organisational integration	The solution should fit existing work practices and provide an accessible interaction channel for intended users.
Cost and resource requirements	Development effort, platform consumption, infrastructure, model usage, and ongoing support should remain proportionate to the expected organisational value.
Scalability	The architecture should support future knowledge domains and workflows without creating uncontrolled complexity.
Evaluation capability	The approach should allow response quality, grounding, unsupported claims, cost, and practical usefulness to be assessed.

As shown in Table 6.1, the criteria included both organisational and technical considerations. This reflected the finding that the suitability of an AI-supported EA artefact could not be determined through model performance alone. A technically capable solution could still be unsuitable if its knowledge sources were difficult to govern, its responses could not be traced to approved material, or its continued maintenance required disproportionate organisational effort.

The technical evaluation also considered multi-hop or relationship-aware reasoning because some EA questions required information to be connected across more than one document or knowledge element. The architecture-evaluation material explicitly identified this requirement and compared Copilot Studio, custom RAG, open-source RAG, and agentic workflows against the organisational context and EA guardrails.

The criteria in Table 6.1 were used to structure the comparative assessment presented in Section 6.4. The later architecture-assessment table should therefore be understood as the application of these criteria rather than as a second independent criteria set.

6.4 Architecture Selection

The comparison did not identify one universally superior architecture. Instead, each approach provided a different balance between implementation effort, organisational integration, retrieval control, technical flexibility, and operational responsibility. The objective was therefore not to identify the technically most advanced solution, but to determine which approach was most appropriate for the selected pilot domain and the requirements established in Chapter 5.

As shown in Table 6.2, Microsoft Copilot Studio demonstrated the strongest fit for a bounded Enterprise Architecture guidance MVP. The platform aligned with the existing Microsoft ecosystem, supported SharePoint-based knowledge sources, enabled guided conversational workflows, and provided a practical path to deployment within the organisational environment. Internal guidance also positions Copilot Studio as suitable for process assistants, knowledge-grounded support agents, onboarding use cases, and structured support activities.

The custom Retrieval-Augmented Generation (RAG) approach offered substantially greater control over retrieval, chunking, metadata enrichment, search strategy, model selection, and evaluation. This made it particularly suitable for use cases requiring stronger grounding, controlled experimentation, relationship-aware reasoning, and more advanced retrieval behaviour. However, these advantages were accompanied by increased engineering effort, operational complexity, and governance responsibilities. Microsoft guidance similarly highlights that enterprise RAG solutions require explicit decisions regarding retrieval design, content preparation, access control, evaluation, and orchestration.

The open-source LLM approach provided the highest degree of flexibility over individual components, models, and deployment architecture. However, it also required the organisation to assume responsibility for a significantly larger technical and governance stack. Within the scope of this thesis and the readiness of the selected pilot domain, the approach was therefore not considered suitable for initial implementation.

Table 6.2 illustrates that the three alternatives occupy different positions with respect to implementation effort, governance alignment, retrieval control, and long-term flexibility. Rather than supporting a binary technology decision, the comparison suggested a staged approach in which different technologies could be applied at different levels of maturity.

The architecture decision therefore distinguished between the initial MVP and the longer-term target direction. For the initial MVP, Microsoft Copilot Studio was selected as the primary interaction and workflow platform. This decision was based on the readiness of SharePoint-based Enterprise Architecture knowledge sources, the suitability of Start Assurance as a structured guidance process, the need for a familiar user experience, and the importance of demonstrating a working artefact within the bounded timeframe of the study. Copilot Studio also enabled guided workflows, process support, and integration with existing Microsoft knowledge repositories without requiring extensive custom development. For the longer-term direction, custom retrieval using Microsoft Foundry was retained as the preferred extension path. This approach becomes increasingly relevant when stronger retrieval control, vector or graph-based retrieval, query classification, systematic model comparison, and explicit evaluation are required. Internal technical investigations similarly concluded that vector and graph retrieval architectures provide greater control over retrieval quality and relationship-aware reasoning than low-code retrieval alone, particularly for complex Enterprise Architecture knowledge domains.

The selected design was therefore not based on a permanent choice between low-code and custom AI. Instead, it adopted a staged architecture:

1. Use Copilot Studio for the governed, bounded, and workflow-oriented MVP.

2. Improve source quality, ownership structures, and document readiness before expanding scope.

3. Evaluate vector and graph-based retrieval for domains requiring stronger retrieval control.

4. Introduce custom Microsoft Foundry services only where the additional complexity is justified by measurable value.

5. Retain human review for questions involving interpretation, exceptions, governance decisions, or formal architecture judgement.

Table 6.2. Comparative Assessment of Candidate Solution Approaches

Dimension	**Copilot Studio**	**Custom RAG in Microsoft Foundry**	**Open-source LLM architecture**
Initial implementation effort	Lower	Higher	Higher
Fit with Microsoft 365-based pilot knowledge	Strong	Strong with additional configuration	Dependent on custom integration
Retrieval and reasoning control	Limited	High	High
Workflow guidance	Strong	Requires custom orchestration	Requires custom orchestration
Organisational governance fit	Strong within the existing Microsoft environment	Strong if Azure governance is established	Requires a dedicated governance and operating model
Security and access-control alignment	Supported through the existing Microsoft environment	Requires configuration through the Azure environment	Requires dedicated security configuration and operational responsibility
Maintenance burden	Lower for a bounded agent	Medium	Higher
Suitability for the initial MVP	High	Medium	Low
Suitability for advanced future use cases	Medium	High	Conditional

***Note**: The terms lower, medium, higher, limited, strong, and conditional represent comparative assessments within the case context. They are not standardised measurements or universally applicable evaluations of the respective technologies.*

6.5 Proposed Artefact Architecture

Based on the stakeholder requirements, architecture evaluation, and MVP prioritisation activities, a Microsoft Copilot Studio-based architecture was selected as the proposed artefact design. The architecture was designed to improve the accessibility of Enterprise Architecture knowledge while minimising implementation complexity and maintenance effort. The solution leverages the existing Microsoft 365 ecosystem and organisational knowledge repositories to provide stakeholders with a conversational interface for accessing architectural information, guidance, and documentation.

Rather than introducing new infrastructure or standalone knowledge repositories, the proposed architecture builds upon existing organisational assets. This approach aligns with the objective of delivering immediate business value while maintaining compliance with organisational governance and security requirements. The architecture enables users to interact with Enterprise Architecture knowledge through natural language queries while retrieving information from approved organisational sources.

6.5.1 Architecture Overview

The proposed artefact consists of four primary components: the user interaction layer, the AI orchestration layer, the knowledge source layer, and the feedback and governance layer. Together, these components support the retrieval, interpretation, and delivery of Enterprise Architecture knowledge in a controlled and traceable manner.

The interaction begins when a user submits a natural language query through the Copilot interface. The query is processed by Copilot Studio, which orchestrates the retrieval process and accesses approved Enterprise Architecture knowledge sources. Retrieved information is then synthesised into a response and presented to the user together with references to the underlying knowledge sources where applicable. User feedback can subsequently be captured and used to support future improvements.

6.5.2 User Interaction Layer

The user interaction layer serves as the primary point of engagement between stakeholders and the artefact. Rather than introducing a standalone application, the proposed solution leverages Microsoft Teams and Microsoft 365 Copilot as the primary interaction channels. This approach allows stakeholders to access Enterprise Architecture guidance within tools that are already integrated into their daily work environment, reducing the need to navigate multiple repositories, websites, or documentation structures. Copilot Studio agents can be published and consumed through Microsoft Teams and Microsoft 365 Copilot, providing a familiar and accessible user experience.

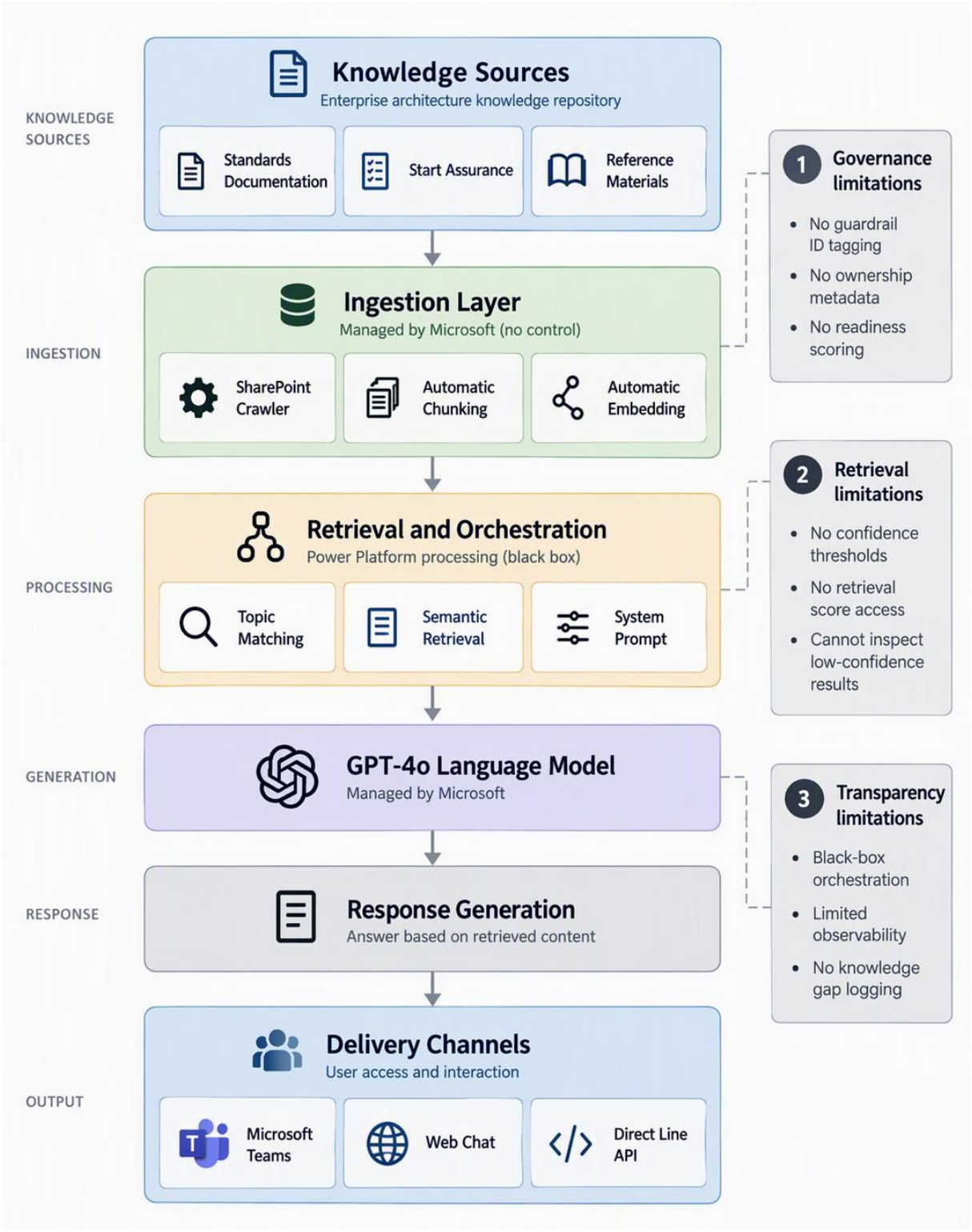


Figure 6.1. Proposed Artefact Architecture with Baseline Microsoft Copilot

6.5.3 Knowledge Retrieval Layer

The knowledge retrieval layer is responsible for identifying and retrieving relevant information from approved organisational knowledge sources. Within the proposed MVP architecture, this capability is provided through Copilot Studio knowledge sources connected to Microsoft 365 repositories, particularly SharePoint-based Enterprise Architecture content. Microsoft documentation describes these knowledge sources as mechanisms for grounding responses using organisational data while respecting existing access permissions.

A key design principle is source grounding. Responses should be traceable to approved organisational knowledge assets to improve transparency, user trust, and answer reliability. This requirement emerged consistently throughout the interviews and workshop, where participants highlighted the importance of understanding where information originates and who is responsible for maintaining it.

6.6 Knowledge Source Design

Data preparation was treated as a central design activity rather than a purely technical implementation task. During the prototype development, a dedicated SharePoint repository was created specifically for the artefact. The repository was curated by removing duplicate files, outdated documents, and content that was no longer considered authoritative. The objective was to establish a cleaner and more reliable knowledge base that could support retrieval and grounding more effectively than the existing distributed document landscape. This approach was informed by both the document-readiness assessment and the prototype experiments, which indicated that curated and well-maintained knowledge sources produced more reliable responses than broad collections of heterogeneous content.

The study therefore emphasised identifying high-value document sets, assigning clear ownership where possible, applying consistent naming conventions, and structuring content in a manner that improves retrievability. By consolidating and cleaning selected Enterprise Architecture knowledge assets in a dedicated repository, the artefact was able to retrieve smaller and more meaningful knowledge fragments while preserving contextual metadata. This was considered important not only for improving answer quality and reducing retrieval noise, but also for supporting transparency, governance, and future maintenance of the knowledge base.



Figure 6.2. Curated SharePoint Knowledge Repository

6.7 Prototype Design

The prototype was designed as an AI-supported Enterprise Architecture guidance assistant implemented using Microsoft Copilot Studio and deployed through Microsoft Teams and Microsoft 365 Copilot. The purpose of the prototype was not to address every identified Enterprise Architecture use case, but to demonstrate how AI-supported guidance could improve access to Enterprise Architecture knowledge within a controlled and manageable scope. The selected pilot domain focused primarily on document discovery and Start Assurance support, two areas that emerged as high-value, high-frequency, and sufficiently mature during the workshop prioritisation process.

The prototype was designed to demonstrate an end-to-end support flow. The interaction begins when a stakeholder submits a natural language question through Teams or Microsoft 365 Copilot. The request is processed by the Copilot Studio agent, which retrieves information from approved Enterprise Architecture knowledge sources and generates a response grounded in the retrieved content. Depending on the type of request, the artefact may provide direct answers,

suggest relevant documents, explain architectural concepts, guide users through parts of the Start Assurance process, or route users to appropriate experts when further support is required.

Particular emphasis was placed on guided process support rather than simple question answering. During prototype development, structured workflows were created for Start Assurance-related interactions, allowing the artefact to ask follow-up questions, collect information, present relevant guidance, and direct users towards appropriate next steps. This approach was considered more valuable than a traditional search interface because it supports both information retrieval and process-oriented guidance.

To improve the quality of retrieved answers, a dedicated SharePoint repository was created as part of the prototype environment. Duplicate documents, outdated files, and content that was no longer considered authoritative were removed, resulting in a curated knowledge base specifically designed to support the selected pilot use cases. This activity was motivated by earlier workshop findings that highlighted issues relating to fragmented repositories, inconsistent document quality, and knowledge-readiness concerns. A focused and curated repository was therefore considered a necessary prerequisite for improving retrieval accuracy and reducing irrelevant results.

The prototype was intentionally bounded in scope. More advanced use cases such as security architecture support, integration discovery, and capability-to-system mapping were not included in the MVP due to higher complexity, lower information readiness, or stronger dependence on specialist expertise. Instead, the prototype focused on demonstrating practical support for recurring Enterprise Architecture needs while establishing a foundation for future extensions.

6.8 Feedback Loop and Continuous Learning Concept

While the prototype provides access to existing Enterprise Architecture knowledge, the usefulness of the artefact depends on its ability to evolve as organisational knowledge changes. Enterprise Architecture documentation, ownership responsibilities, governance structures, and business capabilities are continuously updated. Consequently, a mechanism is required to identify knowledge gaps, support content maintenance, and improve retrieval quality over time.

The proposed feedback mechanism follows a principle of governed learning rather than autonomous learning. User feedback is used to identify areas where improvements may be required, but feedback does not directly change the behaviour of the artefact or modify the underlying knowledge base. Instead, the feedback is reviewed through existing governance structures to ensure that updates remain aligned with approved Enterprise Architecture guidance. This approach supports continuous improvement while preventing uncontrolled model drift or the introduction of unauthorised information.

The feedback concept includes three complementary mechanisms. First, users can indicate whether a response was helpful or whether additional support was required. Second, unanswered questions and retrieval failures can be logged and reviewed to identify missing knowledge sources or content gaps. Third, recurring user requests can be analysed to identify opportunities for expanding the artefact's capabilities or improving existing workflows.

The objective of the feedback mechanism is not to retrain the underlying model automatically, but to improve the quality, relevance, and reliability of the approved knowledge sources, system instructions, and controlled workflows.

6.9 MVP and Scaling Roadmap

The proposed artefact is intended to be implemented incrementally, beginning with a focused MVP centred on Start Assurance and supporting document discovery. Guardrail interpretation, onboarding, business capability navigation, and governance support remain candidate areas for later expansion, subject to knowledge readiness, validation, and risk.

Following MVP validation, the solution can be expanded to include additional Enterprise Architecture domains such as governance support, onboarding assistance, capability-to-system mapping, and integration discovery. This phased approach enables knowledge quality, governance processes, and user adoption to mature before introducing more advanced functionality.

From a longer-term perspective, more advanced retrieval approaches based on Azure AI Foundry and GraphRAG architectures could be explored to support complex reasoning across interconnected Enterprise Architecture artefacts. Such capabilities may enhance contextual understanding, dependency analysis, and multi-hop reasoning, particularly as the volume and maturity of architectural knowledge assets increase. However, these capabilities were considered beyond the scope of the MVP and represent potential future development opportunities rather than immediate implementation priorities.

7. Evaluation

This chapter presents the formative evaluation of the AI-supported Enterprise Architecture guidance artefact. The evaluation examined whether the prototype demonstrated the intended behaviour within the selected pilot domain and whether its design was considered relevant to the organisational context.

The evaluation concentrated on the Copilot Studio-based MVP described in Chapter 6. Particular attention was given to knowledge discovery, guided Start Assurance support, source grounding, response clarity, and the identification of situations requiring human involvement. The purpose was not to demonstrate error-free or deterministic performance, but to assess whether the artefact could provide useful first-line support within a bounded EA knowledge environment.

The chapter first describes the evaluation approach and demonstration procedure. It then presents the qualitative stakeholder feedback, evaluates the artefact against its design requirements, and discusses its principal strengths and limitations.

7.1 Evaluation Approach

The prototype was evaluated formatively using representative evaluation use cases derived from the empirical investigation. An evaluation use case refers to a realistic EA-related question or support activity used to examine the behaviour of the prototype. The term does not refer to a separate set of use cases or a new category of empirical data.

The selected evaluation use cases reflected the principal pilot capabilities established in Chapter 5:

1. locating relevant EA documents and guidance;
2. answering questions based on the curated knowledge sources;
3. supporting users in understanding the Start Assurance process;
4. guiding users through structured Start Assurance questions;
5. identifying circumstances requiring clarification or EA expert involvement.

The formative character of the evaluation was appropriate to the maturity of the artefact. The system represented a bounded MVP rather than a production-ready enterprise service. Consequently, the evaluation focused on whether the prototype demonstrated the intended functionality and whether the design assumptions appeared practically relevant within the case context.

The prototype responses were considered in relation to the curated organisational sources and the professional knowledge of EA team members. The assessment focused on whether the artefact retrieved relevant information, represented the available guidance appropriately, remained within its intended scope, and supported an appropriate next step. Questions involving incomplete evidence, exceptions, or architectural judgement were also used to examine whether the artefact required clarification or human escalation.

The criteria reflect the design requirements derived from the interviews, document analysis, and workshop. They were used to support consistent discussion of the prototype rather than to produce a statistically validated performance score.

Table 7.1. Criteria Used in the Formative Prototype Evaluation

Evaluation criterion	**Evaluation question**
Relevance	Did the response address the user's question?
Source grounding	Was the response supported by the curated EA knowledge sources?
Clarity	Was the response understandable to the intended user?
Practical usefulness	Did the response help the user understand an appropriate next step?
Scope adherence	Did the artefact avoid unsupported or out-of-scope guidance?
Process guidance	Did the controlled workflow guide the user through the relevant Start Assurance steps?
Handling of uncertainty	Did the artefact avoid unsupported certainty when the available evidence was insufficient?
Human escalation	Did the interaction preserve the need for expert involvement in judgement-intensive situations?

7.1.1 Evaluation of a Probabilistic Language Model

The evaluation also considered the inherent characteristics of the underlying Large Language Model. An LLM generates text by estimating probability distributions over possible output tokens and producing tokens sequentially. Its response is therefore probabilistically generated rather than returned as a fixed record from a deterministic information system.

This distinction does not make accuracy unimportant. Instead, it influences how precision should be understood in relation to the intended use of the artefact. For routine knowledge discovery, the prototype should identify relevant sources, represent their content accurately, and avoid unsupported statements. For questions involving architectural exceptions, trade-offs, or decisions, the desired behaviour may be to present relevant information and direct the user to an expert rather than produce a definitive recommendation.

Grounding was used to reduce dependence on the model's pre-trained knowledge. Retrieval-Augmented Generation allows the response to be conditioned on information retrieved from selected organisational sources. However, recent reviews show that grounding does not guarantee factual correctness. Retrieval may return incomplete, irrelevant, or conflicting information, and the generative model may omit or misinterpret otherwise relevant evidence.

The NIST Generative AI Profile similarly treats false or misleading generative output as a risk that must be governed and evaluated throughout the AI lifecycle. Trustworthiness should therefore be incorporated into system design and evaluation rather than inferred from the fluency of a generated response.

Accordingly, the evaluation did not seek to demonstrate that the prototype could provide deterministic or error-free answers. It examined whether curated knowledge sources, retrieval grounding, controlled workflows, source transparency, and human escalation made the probabilistic model sufficiently dependable for bounded first-line EA support.

7.2 Prototype Demonstration and Feedback Collection

The prototype was evaluated through structured online demonstrations and subsequent review discussions with available members of the Enterprise Architecture team. The evaluation examined two forms of prototype behaviour: retrieval of information from curated EA sources and controlled guidance through the Start Assurance process. The controlled workflow was demonstrated using representative questions derived from the stakeholder workshop.

A dedicated AI workflow demonstration was conducted on 1 July 2026. The researcher led the session and demonstrated the Copilot Studio-based prototype using selected questions derived from the stakeholder workshop. These questions were chosen because they represented the principal capabilities of the pilot: locating approved EA material, explaining documented

guidance, supporting the Start Assurance process, and recognising situations requiring further involvement from the EA function. The questions were therefore not created specifically to produce favourable results but were drawn from the question bank established during the earlier empirical work.

The demonstration followed a structured sequence. First, the researcher introduced the purpose and boundaries of the prototype and explained that the artefact was intended to provide first-line guidance rather than make formal architectural decisions. Second, the participants were shown how a user could submit a natural-language question and receive information grounded in the curated SharePoint source environment. Third, the controlled Start Assurance workflow was demonstrated, including the collection of user input, validation of responses, branching according to the information provided, and presentation of an appropriate next step. Finally, the session considered cases in which the available sources were insufficient or professional judgement remained necessary.

Two forms of prototype behaviour were examined. The first was knowledge retrieval, where the artefact attempted to locate and explain information from selected EA documents. The second was controlled process guidance, where a predefined Copilot Studio topic guided the user through part of the Start Assurance routine. This distinction was important because the process-guidance capability did not depend entirely on unrestricted language generation. The workflow constrained the interaction through predefined questions, validation rules, and branching logic.

The representative questions used during the demonstration are presented in Table 7.2. They covered Start Assurance, document discovery, guardrail interpretation, and business-capability ownership. The questions represented different levels of complexity. Document discovery primarily examined whether the artefact could locate approved material. The Start Assurance question examined whether the controlled workflow could collect relevant information and provide process guidance. The guardrail question examined whether the artefact could present documented guidance without treating general information as a formal interpretation or exception decision. The capability-ownership question examined whether the artefact could retrieve documented ownership information without inventing relationships that were absent from the connected sources.

For each demonstrated question, the participants were invited to consider whether the response addressed the question, whether its informational basis could be identified, whether it was understandable, and whether it supported an appropriate next step. Participants also commented on whether the response remained within the intended scope of the artefact and whether further involvement from an architect or another specialist was required. Particular attention was given to questions where an apparently fluent response could still be incomplete, outdated, or insufficiently supported.

Table 7.2. Representative Questions Used During the Formative Evaluation

Evaluation area	Representative question	Expected prototype behaviour
Start Assurance	Do I need Start Assurance for this initiative?	Explain the purpose of Start Assurance, collect relevant information, guide the user through the assessment, and indicate whether EA involvement is required
Document discovery	Where can I find the Start Assurance material?	Locate and reference the approved process description, self-assessment material, or supporting template
Guardrail interpretation	Which Enterprise IT Guardrail may apply to this initiative?	Retrieve the relevant documented guidance, explain its general relevance, and preserve human review for contextual interpretation or exceptions
Business capability lookup	Who owns this business capability?	Retrieve documented capability and ownership information without inventing relationships that are absent from the source material

The demonstration revealed that the condition of the knowledge sources directly influenced the resulting responses. Earlier testing using a broader source environment had returned some outdated information. This led to the selection and cleaning of a smaller set of documents for the prototype knowledge repository. Later demonstrations therefore used the curated source environment rather than the complete distributed collection of EA material. The company-facing thesis presentation subsequently reported the same observation: broader source collections produced quick but sometimes outdated answers, whereas the curated source environment produced more reliable responses for the bounded pilot.

The controlled Start Assurance workflow was the clearest demonstration of the distinction between a general chatbot and the proposed artefact. The workflow asked structured questions and followed predefined logic rather than relying on the language model to infer every step independently. The later thesis presentation similarly described the agent as using controlled topic flows to ensure that the required questions and process steps were addressed.

Feedback was provided verbally during the demonstrations and subsequent thesis progress discussions. The researcher documented the comments as qualitative evaluation notes and

compared them with the design requirements defined earlier in the study. The feedback was organised around relevance, source grounding, clarity, practical usefulness, workflow behaviour, handling of uncertainty, scope adherence, and the need for human escalation.

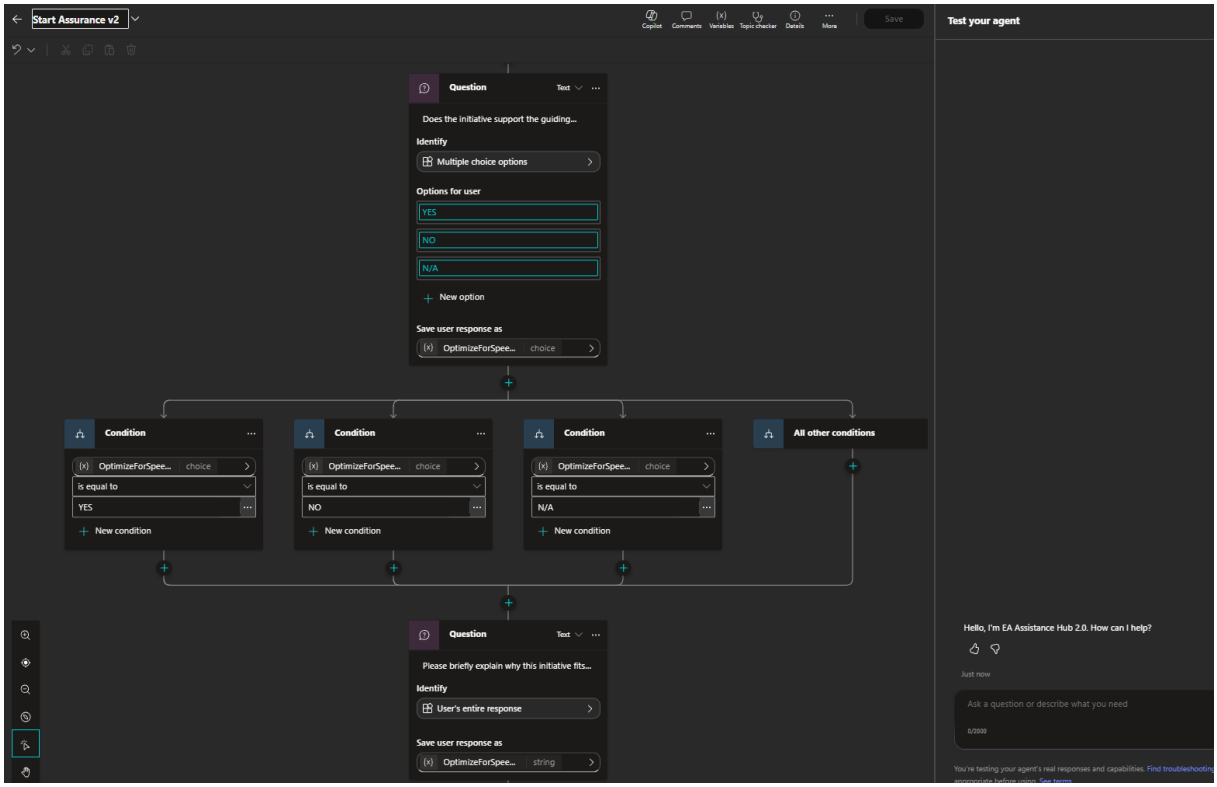


Figure 7.1. Controlled Start Assurance workflow implemented in Microsoft Copilot Studio

As shown in Figure 7.1, the workflow presents a predefined assessment question with the response options *Yes*, *No*, and *N/A*. Each response is stored and evaluated through a corresponding conditional branch before the conversation proceeds to the next question. This structure was intended to maintain the interaction within the defined Start Assurance scope and ensure that predictable process steps were handled through controlled logic rather than repeated open-ended language-model generation.

The evaluation procedure nevertheless provided a structured method for examining whether the prototype reflected the stakeholder-derived requirements and whether it was appropriate for continued development. The findings were recorded against the criteria in Table 7.1 and subsequently summarised in Sections 7.3 and 7.4. The evaluation therefore supported refinement and assessment of the design while preserving a clear distinction between demonstrated prototype behaviour, stakeholder interpretation, and capabilities that remained conceptual.

Feedback was collected verbally during the demonstration and subsequent review discussions. The researcher documented the comments as qualitative evaluation notes and organised them according to the criteria in Table 7.1. The resulting observations concerning source quality, workflow guidance, scope boundaries, confidence communication, and human oversight are presented in Section 7.3.

7.3 Stakeholder Evaluation Results

The feedback collected during the online sessions indicated that the prototype was considered relevant for recurring and bounded EA support needs. The principal value identified in the demonstrations concerned the ability to combine access to EA information with structured conversational guidance.

The feedback was consistent with retaining Copilot Studio as the MVP platform. The platform enabled the artefact to combine retrieval from selected sources with controlled dialogue and workflow logic. This was considered appropriate for recurring questions and defined processes where the expected interaction could be bounded in advance.

At the same time, the evaluation identified limitations in using the prototype for questions requiring extensive reasoning across several documents or architectural relationships. Later technical discussions therefore retained vector and graph-based retrieval as possible alternatives for more advanced knowledge requirements. The August progress discussion explicitly positioned Copilot Studio as the MVP while retaining knowledge-graph and vector-based approaches for comparative investigation.

Feedback also highlighted the importance of knowledge-source quality. The ability of the prototype to provide a relevant response depended on whether the necessary content existed, was current, and had been included in the curated repository. This observation was consistent with the earlier document-readiness assessment. Retrieval could improve access to available knowledge, but it could not independently compensate for absent, outdated, or ambiguous sources.

A further result concerned the treatment of confidence. Discussions explored the possibility of using indicators to communicate how strongly a response was grounded in the available sources. However, the discussions also recognised that a confidence label should not be interpreted as proof of factual correctness. A response could appear plausible while its supporting citation or relationship to the source remained incomplete. Source inspection and human judgement therefore remained necessary for consequential uses. The evaluation consequently supported a differentiated view of automation. The prototype was considered suitable for:

- locating approved EA material;
- answering bounded questions supported by curated sources;
- explaining documented process information;
- guiding users through structured Start Assurance preparation;
- collecting information before engagement with the EA function;
- directing users towards further support.

Human involvement remained necessary where:

- the available documents did not provide sufficient evidence;
- sources appeared inconsistent or outdated;
- the user requested an exception to an established guardrail;
- the question required an architectural trade-off to be assessed;
- the output could influence a formal governance or architecture decision;
- additional organisational context was necessary;
- the artefact could not remain within an approved knowledge scope.

The results therefore did not support the replacement of Enterprise Architects. Instead, the evaluation indicated that the prototype could augment EA practice by supporting recurring first-line interactions and preserving expert capacity for questions requiring professional judgement.

7.4 Evaluation Against Design Requirements

The prototype was evaluated against the design requirements established through the empirical investigation. Table 7.3 summarises the extent to which each requirement was demonstrated during the formative evaluation.

The table distinguishes between requirements demonstrated directly through the MVP and those that remained partly demonstrated or conceptual. This distinction is important because the evaluation did not test the artefact through organisation-wide deployment.

Natural-language access and controlled workflow guidance were the most clearly demonstrated

capabilities. The prototype showed that users could interact with EA knowledge without manually navigating the underlying repository structure and could be guided through a structured process.

Source transparency and contextual explanation were only partly demonstrated. Although responses could be grounded in curated content, the probabilistic generation process meant that the relationship between the source and the generated statement still required verification. Similarly, contextual explanations were strongest where the process and expected answer had been bounded in advance.

The evaluation also confirmed that human oversight was not an optional feature. It represented a necessary design requirement for controlling uncertainty and preserving architectural accountability. The requirement was therefore retained in the proposed governance model.

Scalability was not established empirically. The prototype demonstrated a possible path for adding further workflows and knowledge domains, but the study did not deploy or evaluate the artefact across the wider organisation. Claims concerning organisation-wide scalability are consequently presented as roadmap considerations rather than evaluation findings.

Table 7.3. Formative evaluation against the artefact design requirements

Design requirement	**Evaluation outcome**	**Assessment**
Natural-language knowledge discovery	Users could submit EA-related questions through the conversational interface	Demonstrated within the curated scope
Retrieval from approved sources	The artefact used selected EA content in the SharePoint-based knowledge environment	Demonstrated, but dependent on source quality
Contextual explanation	The artefact could explain retrieved information in relation to bounded questions	Partly demonstrated
Structured Start Assurance guidance	Controlled workflows supported input collection, validation, and branching	Demonstrated in the prototype
Source transparency	Grounding connected answers to selected organisational knowledge	Partly demonstrated; source verification remained necessary

First-line EA support	Routine information and preparation activities could be supported before expert engagement	Supported by qualitative feedback
Human oversight and escalation	The need for continued human involvement was identified for higher-risk questions	Confirmed as necessary
Governed feedback	Verbal feedback and response issues could be captured for review	Conceptually demonstrated; no autonomous learning loop
Future scalability	Additional workflows and retrieval approaches were technically conceivable	Not established through production deployment

7.5 Strengths of the Prototype

The first strength of the prototype was its alignment with a recurring organisational need. The artefact did not begin from a general assumption that EA required a chatbot. Its pilot scope was derived from stakeholder interviews, use-case analysis, document assessment, and workshop prioritisation. This established traceability between the organisational problem and the implemented design.

A second strength was the combination of knowledge retrieval and controlled process guidance. Document discovery alone could help users locate information, but it did not address the process-oriented nature of Start Assurance. The controlled workflow enabled the prototype to request relevant information, follow predefined logic, and guide the user through a bounded sequence.

A third strength concerned organisational integration. The interaction layer was based on Microsoft Teams and Microsoft 365 Copilot, while Copilot Studio provided the agent and workflow capability. This reduced the need for a separate user-facing application and provided a familiar interaction environment for the pilot.

A fourth strength was the use of a curated knowledge repository. Duplicate, outdated, or unsuitable material was removed from the pilot source collection, and the remaining documents were selected according to their relevance to the intended use cases. This reduced some of the retrieval noise associated with using the entire distributed EA document environment.

A fifth strength was the explicit recognition of governance and human oversight. The artefact was framed as first-line guidance rather than as an autonomous architectural authority.

Questions involving uncertainty, exceptions, or formal decisions remained subject to human involvement.

Finally, the staged architecture decision supported both immediate implementation and future development. Copilot Studio provided a practical MVP path, while custom retrieval in Microsoft Foundry remained available for future use cases requiring greater retrieval control or relationship-aware reasoning.

7.6 Limitations of the Prototype and Evaluation

The principal limitation was the dependence on source quality. The artefact could only retrieve knowledge represented in the connected source environment. Missing, outdated, ambiguous, or insufficiently structured information reduced the reliability and completeness of the resulting response.

A second limitation concerned the probabilistic nature of the language model. Grounding connected the prototype to curated EA sources, but it did not transform the underlying model into a deterministic information system. The generated response could still omit relevant details, interpret evidence incorrectly, or express a conclusion with greater certainty than the source justified.

A third limitation concerned questions requiring multi-hop or relationship-aware reasoning. Copilot Studio provided a suitable basis for the bounded MVP, but the study identified limitations where responses required several relationships to be connected across documents or architectural entities. Vector and graph-based retrieval were explored as possible future approaches, but their comparative performance was not established through a complete controlled evaluation.

A fourth limitation related to the feedback mechanism. Feedback was collected verbally during online sessions, and possible improvements could be identified from unresolved questions or stakeholder comments. However, the prototype did not include an autonomous learning mechanism. Updating knowledge or workflow behaviour required governed review and modification. This limitation was intentional because uncontrolled learning could weaken source authority and accountability.

A fifth limitation concerned the evaluation procedure. The evaluation relied on online demonstrations and qualitative feedback from members of the EA team. It did not involve a formal usability instrument, blinded assessment, controlled comparison group, or statistically validated scoring procedure. The results therefore provide evidence of practical relevance and design suitability but do not establish quantitative improvements in efficiency, accuracy, or user adoption.

A sixth limitation was the bounded number of evaluation use cases and knowledge sources. The prototype concentrated on Start Assurance and supporting document discovery. The findings cannot automatically be generalised to more complex EA domains such as security analysis, integration discovery, capability-to-system mapping, or formal architecture decision-making.

Finally, continued use of the artefact would require clear organisational ownership. The pilot repository, workflow logic, instructions, access configuration, feedback review, and validation practices would need to be maintained after the completion of the thesis. Without such an operating model, the artefact could gradually retrieve outdated guidance despite demonstrating useful behaviour during the prototype period.

Taken together, these limitations indicate that the prototype should be interpreted as a validated design direction rather than a completed enterprise service. The evaluation demonstrated the relevance of grounded retrieval and guided workflows for bounded first-line EA support, while confirming that source governance, further testing, and human architectural judgement remain necessary.

8. Discussion

8.1 Discussion of RQ1: EA Knowledge-Access Challenges

The findings indicate that the primary challenges associated with Enterprise Architecture (EA) knowledge accessibility are not caused by a lack of knowledge, but by difficulties related to discovery, interpretation, ownership, and reuse. Enterprise Architecture information was found to be distributed across multiple repositories, documents, presentations, process descriptions, and specialist knowledge domains. While substantial knowledge assets existed, stakeholders frequently experienced difficulties identifying the most relevant source, understanding the context behind architectural guidance, and determining how specific principles or guardrails should be applied in practice.

The results support the knowledge management perspective discussed in Chapter 2. Although much of the architectural knowledge existed in explicit form, its effective use often depended on contextual understanding held by individual architects and subject-matter experts. As a result, many stakeholders relied on Enterprise Architecture experts to act as intermediaries between formal documentation and practical decision-making. This dependence created bottlenecks and limited the scalability of support. The findings therefore suggest that the challenge is not simply information storage, but knowledge accessibility and interpretation.

These findings are consistent with knowledge-management literature distinguishing explicit knowledge from tacit and context-dependent understanding. The EA documents, guardrails, capability maps, and process descriptions represented explicit organisational knowledge, but their practical application often depended on experience held by architects and other specialists (Nonaka and Takeuchi 1995; Nonaka and von Krogh 2009). This helps explain why the presence of formal documentation did not remove stakeholders' dependence on individual experts. Davenport and Prusak (1998) similarly argue that organisational knowledge derives value from context and application rather than from storage alone. More recent EA research identifies fragmented knowledge silos, loss of tacit knowledge, semantic under specification, and governance misalignment as continuing barriers to effective knowledge use (Toumi, Fosso Wamba and Hafsi 2025). The findings therefore suggest that the problem was not an absence of EA content, but limited access to knowledge that was current, authoritative, and sufficiently contextualised for practical use.

8.2 Discussion of RQ2: Stakeholder Needs and Design Requirements

The empirical investigation identified several recurring support needs that were considered suitable for AI-assisted guidance. Stakeholders consistently emphasised knowledge discovery, onboarding, guardrail interpretation, Start Assurance support, business capability navigation, and governance-related questions as areas where AI could create value. These findings align with recent literature suggesting that AI assistants are particularly valuable when knowledge is distributed across multiple repositories and users require contextual support rather than simple search functionality.

The workshop further demonstrated that not every identified use case was equally suitable for implementation. The prioritisation exercise revealed that value, frequency, readiness, and risk must be considered alongside technical feasibility. Questions relating to document discovery and Start Assurance received strong support because they occurred frequently, relied on identifiable knowledge sources, and could be supported through structured guidance. In contrast, security-related support and advanced capability mapping were considered more complex due to higher risk and lower information readiness.

The identified requirements also show that stakeholders needed more than a conventional search function. Enterprise search can improve information discovery, but its effectiveness depends on content quality, metadata, organisational terminology, and continuing management of the search environment (White 2015). The participants' requests for understandable explanations, conversational clarification, and source transparency indicate a need to support interpretation as well as retrieval. This is consistent with research on conversational agents, which identifies communication quality, trust, adoption, and organisational value as important considerations alongside technical language-processing capability (Mariani, Hashemi and Wirtz 2023). The requirements for approved sources, visible ownership, human escalation, and bounded interaction further demonstrate that stakeholder trust could not be established through conversational fluency alone. In this case, trust depended on whether users could recognise the source of the response and understand when an EA specialist remained responsible for further interpretation.

8.3 Discussion of RQ3: Suitability of the Selected AI Approach

The comparison of candidate architectures demonstrated that different AI approaches address different organisational needs. Microsoft Copilot Studio provided strong alignment with the existing Microsoft ecosystem, low implementation effort, and rapid deployment potential. It also supported guided conversational workflows and integration with existing SharePoint-based knowledge sources, making it particularly suitable for the selected MVP scope.

Custom Retrieval-Augmented Generation (RAG) architectures provided greater control over retrieval, chunking, metadata management, and evaluation. In particular, graph-oriented retrieval approaches showed promise for supporting relationship-aware reasoning across architectural concepts, capabilities, systems, and governance artefacts. However, these approaches also introduced greater complexity and required additional engineering effort.

The findings suggest that architecture selection should not be driven solely by technical capability. Organisational readiness, governance, maintenance effort, and stakeholder adoption are equally important considerations. The most appropriate architecture depends on the maturity of the knowledge environment and the objectives of the intended solution.

The architecture comparison supports a contingent rather than universally hierarchical view of the alternatives. Retrieval-Augmented Generation was originally proposed as a mechanism for combining the parametric knowledge of a language model with information retrieved from external non-parametric memory (Lewis et al. 2020). This provides a stronger basis for source-grounded, domain-specific responses but does not by itself guarantee reliable output. Lewis et al. (2020) also identify provenance and the updating of model knowledge as important problems in knowledge-intensive generation. Recent reviews similarly show that RAG effectiveness depends on retrieval quality, source preparation, orchestration, and evaluation, while enterprise implementations remain affected by the gap between experimental prototypes and operational deployment (Brown, Roman and Devereux 2025; Karakurt and Akbulut 2026). The selection of Copilot Studio for the bounded MVP therefore does not imply that it was technically superior in every dimension. It indicates that organisational integration, workflow control, implementation effort, and maintainability were more important for the immediate pilot, whereas custom vector or graph-based retrieval remained more appropriate for future use cases requiring greater retrieval control and relationship-aware reasoning.

8.4 Discussion of RQ4: Artefact Design and Evaluation

The Design Science Research process resulted in a proposed AI-supported Enterprise Architecture guidance artefact that combines conversational interaction, knowledge retrieval, structured workflows, source grounding, and governance mechanisms. The artefact was designed as a first-line support mechanism rather than a replacement for architectural expertise.

A key insight from the study is that successful AI-supported guidance depends as much on knowledge quality as on AI capability. The workshop repeatedly highlighted the importance of ownership, validation, document readiness, and source authority. Consequently, the artefact was designed to incorporate governance structures, source ownership, feedback processes, and escalation mechanisms alongside retrieval capabilities

The findings therefore suggest that AI-supported Enterprise Architecture guidance should be viewed as a socio-technical capability rather than as a standalone technology solution. Effective guidance requires appropriate knowledge management practices, governance structures, and organisational ownership in addition to AI functionality.

The development and evaluation of the artefact reflect the central logic of Design Science Research. DSR develops knowledge through the construction and application of artefacts intended to address identified organisational problems (Hevner et al. 2004). In this study, the interviews, document analysis, and workshop established the problem context and design requirements, while prototype development and formative evaluation examined whether the proposed response was suitable within that context. This progression corresponds with the activities proposed by Peffers et al. (2007), particularly problem identification, definition of solution objectives, design and development, demonstration, and evaluation. The contribution is therefore not limited to the Copilot Studio prototype. It also includes the design knowledge generated concerning source preparation, workflow boundaries, knowledge ownership, architecture selection, and human escalation. This positioning is consistent with Gregor and Hevner's (2013) argument that DSR contributions may arise from both the artefact and the knowledge produced through its development and evaluation.

8.5 Discussion of the Working Hypothesis

The working hypothesis proposed that a carefully evaluated AI approach, selected through stakeholder-informed criteria and validated within a pilot domain, could improve access to Enterprise Architecture knowledge by reducing dependence on individual experts and enabling faster and more consistent first-line guidance.

The findings provide partial support for this hypothesis. The prototype demonstrated that curated knowledge sources and controlled conversational workflows could support document discovery and guide users through the initial stages of Start Assurance. Stakeholder feedback also indicated that such support could reduce preparatory and repetitive assistance before the point at which professional architectural judgement was required.

The first underlying assumption concerned the preparation of EA documentation. This assumption was supported by the prototype work, as the use of a curated SharePoint repository reduced the retrieval noise associated with duplicate, outdated, and unrelated material. However, the evaluation did not quantitatively compare retrieval performance against manual search. The study therefore supports the importance of knowledge preparation but does not establish a measured improvement in retrieval efficiency.

The second assumption concerned the evaluation of alternative solution approaches. This assumption was supported. The comparison showed that Copilot Studio was suitable for rapid

implementation of the bounded MVP, while custom RAG in Microsoft Foundry remained a stronger extension path for use cases requiring greater retrieval control and relationship-aware reasoning.

The third assumption concerned feedback and continued knowledge updating. This assumption was only partly supported. The study developed a governed feedback concept and identified how unanswered questions and user feedback could inform updates to sources and workflows. However, a complete operational feedback lifecycle was not evaluated over an extended period.

The working hypothesis is therefore considered **partly supported**. The study provides evidence that AI-supported guidance can improve first-line access to bounded and well-prepared EA knowledge, but it does not demonstrate that human expertise can be removed or that organisation-wide efficiency and consistency have been achieved. Human architectural judgement remains necessary for exceptions, trade-offs, and formal decisions.

8.6 Implications for Enterprise Architecture Practice

The study has several implications for Enterprise Architecture practice.

First, improving access to Enterprise Architecture knowledge may create value without fundamentally changing existing governance structures. The proposed artefact demonstrates how existing documentation, processes, and architectural guidance can be made more accessible through conversational interaction and source-grounded retrieval.

Second, Enterprise Architecture support should be viewed as an ongoing service rather than solely a documentation activity. Stakeholders repeatedly sought contextual explanations, onboarding support, and process guidance. These findings suggest that increasing the accessibility of architectural knowledge may be as important as creating the knowledge itself.

Third, organisations considering AI-supported guidance should prioritise knowledge readiness before focusing on advanced AI capabilities. The workshop demonstrated that source quality, ownership, documentation maturity, and governance directly influence the usefulness of AI-generated responses. Expanding AI access to poorly maintained knowledge sources is unlikely to achieve the intended benefits.

8.7 Implications for Research

The study contributes to the growing body of research at the intersection of Enterprise Architecture, knowledge management, and Artificial Intelligence. Existing literature has examined Enterprise Architecture governance and AI-supported retrieval largely as separate

topics. However, comparatively limited research has explored how AI-supported artefacts can be designed to improve access to Enterprise Architecture knowledge in practice.

The findings suggest that knowledge accessibility represents an underexplored research area within Enterprise Architecture. The study further demonstrates how Design Science Research can be used to connect stakeholder needs, organisational constraints, and emerging AI capabilities in a structured design process.

The research also highlights the importance of considering governance, ownership, and knowledge readiness when investigating enterprise AI applications. These organisational considerations are frequently underrepresented in AI-focused research despite their importance in practical deployment contexts.

8.8 Reflection on Design Science Research

The Design Science Research approach proved suitable for the objectives of this study because the research problem required both understanding and intervention. The study sought not only to investigate Enterprise Architecture knowledge-access challenges but also to develop a purposeful artefact capable of addressing those challenges.

The structured progression from problem identification, requirement extraction, artefact design, evaluation, and future recommendations provided a clear link between empirical findings and design decisions. This traceability strengthened the credibility of the resulting artefact by ensuring that key design choices remained grounded in stakeholder needs and organisational realities.

A further benefit of the DSR approach was its ability to balance practical relevance and academic contribution. The resulting artefact addresses an organisational challenge while simultaneously contributing to broader discussions concerning AI-supported Enterprise Architecture guidance.

8.9 Methodological Reflection

The study employed interviews, document analysis, and workshop to investigate the problem context. The combination of these methods enabled triangulation and provided a richer understanding of the organisational environment than would have been possible through a single method.

The workshop was particularly valuable because they transformed interview findings into concrete questions, ownership structures, document mappings, readiness assessments, and

candidate MVP use cases. This strengthened the connection between stakeholder needs and artefact design.

However, the study also has limitations. The findings are derived from a single organisational context and reflect the specific Enterprise Architecture environment of Sandvik Coromant. Furthermore, the artefact was developed as a bounded MVP rather than a fully operational enterprise-wide solution. These limitations do not invalidate the findings but should be considered when interpreting the results and assessing their transferability.

9. Conclusion

9.1 Summary of Findings

This study investigated how Artificial Intelligence could be used to improve access to Enterprise Architecture knowledge within an industrial context. The research demonstrated that significant architectural knowledge already existed within the organisation but that stakeholders frequently experienced difficulties locating, interpreting, and applying that knowledge.

Through interviews, document analysis, and workshop, the study identified recurring challenges related to knowledge discovery, onboarding, guardrail interpretation, governance understanding, and dependence on experienced architects. These findings informed the design requirements of an AI-supported Enterprise Architecture guidance artefact.

The study subsequently evaluated alternative AI architectures and proposed a Copilot Studio-based MVP supported by curated Enterprise Architecture knowledge sources, governance mechanisms, structured workflows, and source-grounded responses. The design further identified Retrieval-Augmented Generation as a promising future evolution path for more advanced retrieval and reasoning use cases.

9.2 Answers to the Research Questions

RQ1: What challenges affect access to Enterprise Architecture knowledge in the case organisation?

The study found that the principal challenges concern knowledge discoverability, interpretation, distributed repositories, inconsistent ownership visibility, and dependence on Enterprise Architecture experts for contextual guidance.

RQ2: What stakeholder needs and design requirements should inform an AI-supported Enterprise Architecture guidance artefact?

Stakeholders required an artefact that could provide natural-language access to approved EA knowledge, simplify document discovery, explain established guidance, and support structured processes such as Start Assurance. The principal design requirements were source-grounded responses, contextual clarification, source transparency, visible knowledge ownership, maintainable sources, human oversight, and escalation where the available evidence was insufficient or professional judgement was required.

RQ3: Which AI solution approach is most suitable for the selected pilot when evaluated against the identified organisational and technical criteria?

No architecture was universally superior. For the selected MVP scope, Microsoft Copilot Studio provided the strongest balance between organisational fit, governance, maintainability, and deployment effort. Custom RAG architectures offered greater retrieval control and reasoning capabilities but required additional engineering effort and operational maturity.

RQ4: How can an AI-supported Enterprise Architecture guidance artefact be designed and evaluated within the case organisation?

The study demonstrated that such an artefact can be designed by combining conversational interaction, source-grounded retrieval, guided workflows, governance mechanisms, ownership structures, and continuous feedback processes.

9.3 Contributions

9.3.1 Practical Contribution

The practical contribution of the study is a stakeholder-informed design for an AI-supported Enterprise Architecture guidance artefact. The proposed solution provides a structured approach to improving Enterprise Architecture knowledge accessibility while maintaining governance, ownership, and expert oversight.

The study also provides a practical roadmap for organisations seeking to evaluate AI-supported guidance solutions within Enterprise Architecture environments.

9.3.2 Research Contribution

The research contributes to the emerging intersection of Enterprise Architecture, knowledge management, and Artificial Intelligence. It extends understanding of how AI-supported guidance can be applied within Enterprise Architecture contexts and demonstrates how Design Science Research can be used to develop and evaluate such artefacts.

The study further contributes knowledge concerning the importance of knowledge readiness, governance, ownership, and organisational constraints in the design of enterprise AI solutions.

9.4 Limitations

Several limitations should be acknowledged.

First, the study was conducted within a single organisational context, limiting the direct transferability of findings to other organisations. Second, the prototype focused on a bounded MVP rather than a fully operational enterprise-wide implementation. Third, the evaluation emphasised organisational relevance and stakeholder validation rather than large-scale quantitative performance testing.

Finally, the technical comparison focused on a selected set of architectures within the constraints of a master's thesis and does not represent an exhaustive evaluation of all available Enterprise AI approaches.

9.5 Future Research

Several opportunities for future research emerge from this study.

Future work could evaluate the artefact through long-term deployment and usage analysis to investigate adoption behaviour, organisational impact, and evolving support needs. Additional studies could examine the effectiveness of graph-based retrieval and knowledge graph approaches for Enterprise Architecture guidance and compare their performance with traditional vector retrieval strategies.

Further research could also investigate multi-agent architectures, workflow orchestration, and automated governance support within Enterprise Architecture environments. Finally, cross-organisational studies would help determine the extent to which the findings can be generalised to other industries and Enterprise Architecture contexts.

Acknowledgement of AI Assisted Tools:

In the preparation of this thesis, AI-assisted tools were used to support document comprehension and language refinement.

NotebookLM was used to assist with the review and organisation of academic source material. Microsoft 365 Copilot was used for grammar correction, sentence restructuring, readability improvement, and refinement of academic expression.

The tools were not used to make research-design decisions or independently generate empirical findings. All data collection, analysis, interpretation, artefact-design decisions, and conclusions are the author's own work. The author assumes full responsibility for the accuracy and integrity of the thesis.

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